# Who Gets a Place When Applicants Follow a Shortlist?

Shortlist design in New York rental simulations and Chilean school admissions

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A school recommendation can predict where a pupil will enroll and still make it harder for someone to get a seat. We reconstruct Chile's regular-round admissions mechanism and reproduce the initial course or no-offer outcome for all 2,044,649 applicants in four cohorts. Among 1,969,120 records linked to enrollment, using submitted rank in an offer predictor puts the school later attended on 4,638 more menus. Clearing those menus places 1,670 fewer pupils. Pupils who originally received no offer account for 1,270 of the net decline. Appending omitted courses actually named again in the complementary round leaves 1,653 fewer linked pupils placed. Field evidence motivates additional tests of who searches. Holding the number of searchers fixed, their identities alone can reverse which rule better covers the school later attended. How much predictive value could a platform retain while protecting an earlier allocation? For fixed priorities and an explicit set of draws, preserving each baseline as a stable allocation permits exact row-by-row optimization. Allowing even a fixed one percent of draws to fail makes the decision problem NP-complete. In three cohorts, a designer given the April enrollment labels can gain at most 314 of the unrestricted switch's 3,590 net additional hits while preserving every applicant's baseline course and seat type in 1,024 draws per cohort. We construct menus attaining 314 and re-clear all 3,072 cohort–draws under Chile's seat and family rules. The pooled-seat model gives 278. After tentative offers, we construct chains of voluntary moves to vacancies. They recover the original placement count in every supply-enriched New York rental simulation. Both settings illustrate what a shortlist can change when the set of options is fixed.

## 1 Introduction

A platform can score a school recommendation by asking whether it included the school a pupil later attended. A family also needs to know whether it will get a seat when others use the same service. We compared two rules trained on 2019 Chilean admissions offers. Both select up to five courses from the family's submitted list and keep any guaranteed continuity course. One also uses the family's submitted rank as a predictor. We evaluated the two rules on four later national cohorts.

We first reconstructed Chile's initial offers, including schools' reserved seats, pupils' guaranteed return to their current school and the effect of an older sibling's assignment. The public course or no-offer record agrees in every case. On the hypothetical menus, using rank covered 4,638 more April schools and placed 1,670 fewer of the same 1,969,120 linked applicants. That is 2.36 more schools covered and 0.85 fewer placements per 1,000. The 145,450 linked pupils who had received no initial offer accounted for 76.0% of the net placement decline. They also lost 43 April-school hits. Against the simpler alternative of keeping the family's first five discretionary choices, the rank-informed predictor gained both hits and placements in every year. The choice of benchmark matters. We also test what happens when families pursue schools beyond the display. Reapplication records supply actual school identities. A separate field experiment motivates response rates. We can bound the coverage comparison for any common set of responders and re-clear the market when the bound reverses its sign.

Which losses come from a pupil losing access to their old course, and which require a change elsewhere? We give a path that starts at a lost seat and identifies a distinct applicant whose two assignments switch order between the two policies. The result allows both the available options

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and the order of application to change. In the fixed-priority Chilean benchmark, 1,834 of the 2,527 pupils who lost a simulated placement still had their former course on their new list. Each of these losses has a witness on another applicant's changed list. The paths locate a necessary disagreement between the two allocations. They do not predict the effect of changing a single witness's list in isolation.

The original motivation came from rental search. Can a service uncover an affordable apartment a household would otherwise miss? In ordinary New York draws, 5,425 of 6,617 lower-income renter instances had no feasible listing at all. The ads and survey budgets let us vary the supply households can afford. They leave household rankings, applications and leases unobserved. Chile supplies submitted preferences, offers and enrollment, along with a protected seat that can make a five-course menu different from a simple cutoff. We use a common first-five benchmark to compare the mechanisms across settings. We also let people recover omitted options in different orders. No Chilean family received either hypothetical menu, and centralized deferred acceptance in the rental study is a model of competition.

We study two decisions that a platform could take at different times. Before priorities are drawn, it can test a menu against several possible lotteries. A finite list gives an exact computational guarantee and lets us measure how much of that guarantee survives a new draw. After tentative offers exist, the platform can ask who would prefer to move to a vacancy and who could use the freed seat. In Chile, allowing every such move, even without enforcing school priorities, leaves most of the placement deficit intact. These bounds on repair give the two studies a common question: how many people can curation help when the available options stay fixed?

## 2 What each experiment measures

Let $E$ be the bipartite graph of pairs treated as acceptable in the replay. It contains the retained regular-round reports in Chile and the modeled feasible apartments in New York. A displayed graph $D \subseteq E$ lists the programs an applicant can pursue in the binding-menu replay. Unless stated otherwise, proposals follow the original order on each displayed list. Section 9 also varies this order. Each applicant strictly orders acceptable programs above being unplaced. The fixed-priority benchmark used for the path and vacancy-chain results has the following structure. A program has a fixed capacity and a strict priority ordering over applicants. It holds the highest-priority acceptable applicants who have proposed, up to its capacity. In the rental simulations each listing has one place. Let $M = \mathrm{DA}(E)$ be the full-list allocation and $N_D = \mathrm{DA}(D)$ the policy allocation; an unplaced applicant receives $\perp$. Stability means that no displayed pair would both prefer to match, taking account of vacancies and priorities. DA on a fixed graph is polynomial-time and applicant-optimal among its stable allocations [Gale and Shapley, 1962, Roth, 1986].

For applicant $i$, define

$$C_i(D) = \mathbf{1}\{M(i) \neq \perp,\ (i, M(i)) \in D\}, \qquad P_i(D) = \mathbf{1}\{N_D(i) \neq \perp\}.$$

Placement counts any acceptable assignment. It gives no extra credit for a first choice. Section B measures rank separately. Both rates use all applicants. A person without a full-list partner contributes zero to coverage and can still receive a place after the lists change. In Chile we also compute $C_i$ using the historical offer instead of $M(i)$. For policies $A$ and $B$, the ordered pair $(C_i(B) - C_i(A), P_i(B) - P_i(A))$ belongs to a three-by-three table. One cell counts people who newly see their full-list partner yet lose a placement. We use a separate indicator for exact equality of the two personal menus to identify people whose options stayed fixed while competitors' lists changed. Unchanged coverage by itself does not imply that a menu stayed fixed.

Table 1. Two policy comparisons with a common allocation outcome. Each school application cycle is a separate national cohort.

| | New York rentals | Chilean school admissions |
|---|---|---|
| Choices and priorities | Affordability and space rules, modeled tastes and screening | Submitted course rankings, published priorities and within-course tie orders |
| Shortlist change | Uniform five to Haiku five | Rank-blind predictor to rank-informed predictor, five discretionary choices plus entitlement |
| Reference option | Simulated full-list DA unit | Replay program, recorded initial offer, and April enrollment school |
| Evidence | Archived model on development markets; first-five rule also tested on separate ads and households | Frozen 2019 models; four later cohorts for the specified signs |

*Relation to prior work.* Competition between recommended options has a long history. Su et al. [2022] optimize rankings jointly when job seekers apply and employers decide whom to interview. Ashlagi et al. [2020] study recommendations that reduce communication in matching markets. McCauley et al. [2026] link unmatched hospitals under prefix truncation to missed stable partners. Our comparisons permit deletions anywhere on a submitted list and trace who is displaced. We use the classical invariance of matched participants on a fixed graph and results on entry and consistency [Crawford, 1991, Gale and Sotomayor, 1985, Roth, 1986, Sasaki and Toda, 1992].

Aguiar and Hong [2026] compare feasible and stable assignments on restricted graphs using Chilean school data. Arteaga et al. [2022b] measure responses to admission-risk advice. Agte et al. [2024] combine observed search with randomized information, and Allende et al. [2026] model how congestion and changes in school quality affect the returns to information at scale. Neilson [2026] documents movement after the centralized assignment. Munro [2026] studies treatment spillovers in designed markets. Here we hold seats and priorities fixed, compare frozen menus against a later enrollment outcome and trace individual losses after clearing. Liu et al. [2024] audit the housing options language models recommend. Our rental exercise asks what happens to those recommendations when several households compete for the same unit.

## 3 Chilean school admissions

The Chilean school choice system records the courses a family ranks, the school's priority and lottery information, and the initial offer the family actually receives [Correa et al., 2022, Ministerio de Educación de Chile, 2026]. We can ask whether a hypothetical shortlist would have included that offer. To learn who would receive a place on the new lists, we must replay the allocation.

We fitted two ridge logistic offer predictors to the same 483,070 applicants in 2019, using eligible discretionary applicant–course edges as observations. Both use program demand, capacity, congestion, list length, grade, priority indicators and a smoothed program offer rate estimated from 2019 (leave-one-out for training). The second adds log rank, a first-choice indicator and a rank–congestion interaction. We call the first score rank-blind because it omits these features. Both pipelines use rank to truncate at the continuity entitlement and to order proposals. Both break equal fitted scores with the same seeded edge-specific random keys. The 2019 fit uses observed offers as training labels; no evaluation-year offer enters either policy. Evaluation-year demand is computed from the entire submitted list profile. A platform could use this score after reports are collected; advance advice would require a demand forecast. The lottery is used for allocation

only. Each predictor retains up to five discretionary choices. A pupil's guaranteed current-school option remains available outside that quota. The eligible graph retains the report down to this entitlement, including the entitlement itself. In the sequel a full-list partner refers to DA on these eligible regular-round entries. The menu construction and its scores stay fixed when we change the allocation mechanism.

*Replaying the historical offers.* We implemented the published regular-round procedure, which runs deferred acceptance from the highest grade down [Correa et al., 2022, Doğan et al., 2026, Ministerio de Educación de Chile, 2026]. For each school and course, the mechanism distinguishes regular, disadvantaged, special-education and academic seats. Pupils can return to their guaranteed course even if its published vacancies run out. An older sibling's admission can give a younger sibling priority at that school. A separate family application can also move the school up the younger pupil's list. In the historical replay we retain choices below the guaranteed course because family processing can promote them. For the shortlist comparison we keep the exact menus that the two predictors originally produced.

For validation we built the full-report market from the released applications, seat counts, priority records and family links. We saved the predictions, then joined them to the ministry's initial offers. Acceptance decisions and wait lists come later. All 2,044,649 predictions agreed, including all 161,591 no-offer records (Table 2). Doğan et al. [2026, Appendix S7] previously reported exact reconstruction across 2017–25. Our question is whether a recommendation result obtained with a simpler mechanism survives when those rules govern the counterfactual too. Appendix F traces the changes in fit and explains what the public record can verify.

*The first confirmation and its extension.* After examining 2020 and 2021 we recorded two predictions for 2022–23. Giving the 2019 predictor access to submitted rank would cover more historical initial offers and reduce placements. Before downloading 2024 and 2025, we committed to the same comparison in both further cohorts. The specified replay pooled the seats at each course, used fixed priority tiers, and raised the published vacancy count to the number of eligible continuing applicants wherever necessary. The predictions held in all four years. The subsequent institutional reconstruction gives the same directions with different magnitudes. Of 2,635 pupils who lose a placement under the reconstructed rule, 1,843 have exactly the same personal menu under both policies. Another 801 pupils gain a placement. No evaluation-year offer enters the scoring or clearing rule. In 2025 the ministry used a deterministic ordering for the coming school year. We use its recorded tie order for historical and menu comparisons and restrict the random-lottery analyses to 2022–24 [Ministerio de Educación de Chile, 2025, 2026].

Table 2. Validation on original reports and effects of changing the menus. An exact replay agrees with the ministry on the entire assigned course or on receiving no offer. The last three columns compare rank-informed with rank-blind menus. Both placement columns count all applicants.

| | | Exact original | Additional original | Change in placements | |
|---|---|---|---|---|---|
| Year | Applicants | outcomes | offers shown | Pooled seats | Chilean rules |
| 2022 | 570,891 | 570,891 | +1, 143 | −468 | −507 |
| 2023 | 536,353 | 536,353 | +1, 350 | −517 | −509 |
| 2024 | 473,482 | 473,482 | +1, 735 | −481 | −473 |
| 2025 | 463,923 | 463,923 | +1, 324 | −321 | −345 |
| Total | 2,044,649 | 2,044,649 | +5, 552 | −1, 787 | −1, 834 |

For comparison with the original protocol, the following benchmarks use pooled seats. In the 2022 and 2023 cohorts we also reran DA on 48 uniform five-choice displays in each cohort and on the family's first five choices. Relative to the uniform-draw mean, the rank-informed policy covered about 9,235 and 9,740 more full-list partners and placed about 789 and 759 more pupils in 2022 and 2023. Relative to first-five, it placed 2,123 and 2,013 more. The rank-blind predictor placed the most pupils of these four policies in both years. Prioritizing predicted offers improved on simple lists, and adding rank information then moved the two outcomes in opposite directions.

*A market-wide predictor.* Across 2022–24, a rule using the completed applications to simulate unknown priority draws covered more historical offers and placed more pupils than the rank-informed predictor in every 64-draw batch. The placement advantage was small, from 2 to 56 pupils per batch, and each batch gave fewer first choices. When compared with the rank-blind rule it still placed fewer applicants. Appendix K gives the information available to the simulated rule, its tie handling, and the results at smaller simulation budgets. It also reports seat, quota and priority-order sensitivities. Under a larger seat convention the replay fits the recorded assignment less well, although the original policy comparisons retain their signs.

A further rule chooses five courses that together clear at least one simulated admission cutoff as often as possible. Under independent course lotteries it placed more pupils than ranking by partner frequency in all 24 paired comparisons. The rank-blind predictor still placed more. Appendix H reports the cost in first choices and an alternative lottery specification in which most of the gain comes from how ties are broken.

## 4 Did the shortlist contain the school the pupil later attended?

The predictors were trained on offers. We can test them against a different outcome recorded months later by linking each original applicant to the national school register for the following April 30 [Ministerio de Educación de Chile, 2026]. Of 2,044,649 applicant-years, 1,969,120 link to a registered school. A menu scores one hit if any displayed course is at that school. The target is the same historically attended school for both rules. For the placement comparison, we clear the entire cohort under the reconstructed Chilean rules, then count placements among the register-linked applicants.

Table 3. Historical school enrollment predicted and placements under the reconstructed rules. The middle columns count linked applicants whose five-choice menu, including any protected option, contains their April school. The last column is the simulated placement change for those same applicants.

| Application year | Linked | Blind hits | Informed hits | Hit change | Placement change |
|---|---|---|---|---|---|
| 2022 | 549,997 | 444,227 | 445,234 | +1,007 | −441 |
| 2023 | 516,921 | 420,820 | 422,018 | +1,198 | −479 |
| 2024 | 457,231 | 374,906 | 376,291 | +1,385 | −440 |
| 2025 | 444,971 | 365,988 | 367,036 | +1,048 | −310 |
| Total | 1,969,120 | 1,605,941 | 1,610,579 | +4,638 | −1,670 |

*Does the enrollment result simply reproduce the initial offer?* Most of the coverage gain over rank-blind came from pupils who enrolled at their initial offer school (Table 4). Among pupils who enrolled elsewhere, 1,357 gained coverage of their April school and 771 lost it. The net gain of 586 was positive in each cohort. The 145,450 linked pupils without an initial offer lost 43 hits on net under the binding menus. They accounted for 76.0% of the overall net placement decline. These are

Table 4. The same rank-informed menu against two alternatives. Entries are net changes in pupils. A hit means that the menu contains the historical April school. Placement is simulated after clearing the entire cohort. The three historical groups exhaust the linked sample.

| | | Versus rank-blind | | Versus first five | |
|---|---|---|---|---|---|
| Initial offer and April school | Applicants | Hits | Placed | Hits | Placed |
| Attended initial offer school | 1,360,401 | +4,095 | −106 | +7,740 | +6,149 |
| Offered a seat, enrolled elsewhere | 463,269 | +586 | −294 | +1,555 | +4,854 |
| Received no initial offer | 145,450 | −43 | −1,270 | +505 | −2,860 |
| All linked pupils | 1,969,120 | +4,638 | −1,670 | +9,800 | +8,143 |

descriptions of historical groups. Receipt of an offer and the subsequent enrollment decision both occur after the original report.

Against keeping the first five discretionary courses plus the entitlement, the rank-informed rule improved both aggregate measures in every cohort. Even against this simpler comparison it placed fewer pupils in the group with no initial offer. Requiring school, education type and grade to match in the original rank-informed versus rank-blind comparison gives a net gain of 4,487 hits.

*Missing registers.* Could missing registers reverse the school-level result? For an unlinked pupil, the blind rule can gain a hit over the informed rule only if it lists a school absent from the informed menu. Just 3,760 unlinked records meet that condition. The reverse condition holds for 3,757. Allowing each of the 75,529 unlinked records any school or no enrollment gives a full-cohort net gain between $4,638 - 3,760 = 878$ and $4,638 + 3,757 = 8,395$. The lower bound is positive in each year (54, 313, 442 and 69). These are bounds from the two menus, with no assumption about why a register link is missing.

The average conceals a limit precisely where families needed another option. Among the 126,708 linked pupils still without a regular offer after the wait list, only 25,499 (20.12%) attended a school anywhere on their eligible original report. The blind and informed menus contained the eventual school for 25,158 and 25,122, respectively. Even a perfect predictor restricted to that report would miss nearly four in five of their April schools. The complementary round records where some of them searched next.

*Where families searched next.* A submitted list is a record of one search. To see whether families subsequently pursued other schools, we linked the regular and complementary application files by masked pupil identifier within each of the four years. The complementary round allows eligible families to submit another list from schools with vacancies [Ministerio de Educación de Chile, 2024]. We selected everyone whose initial and post-wait-list regular outcomes both recorded no offer. We counted a new school only when its identifier was absent from *all* original entries and the family explicitly listed a course there in a same-grade complementary application. Automatic continuity entries and assignments to a course the family never listed do not count as new explicit offers. Finally we linked the initial outcome in the complementary round to the next school year's April 30 enrollment register by pupil and school [Ministerio de Educación de Chile, 2026, Ministerio de Educación de Chile, 2026].

Of the 141,345 initially and finally unoffered regular applicants, 21.72% subsequently enrolled at a school absent from the original list through this observed sequence. Among 54,401 offered an explicitly added school, 51,956 appeared anywhere in the following April's register and 30,699 appeared at that school (56.43% of all 54,401). Another 21,257 appeared at a different school and 2,445

Table 5. Observed choices after the regular round, in application records. Each column is a subset of the column to its left. Enrollment is at the new school where the complementary round offered the listed course.

| Year | No regular offer | Reapplied[a] | Listed new school | Offered it | Enrolled there |
|---|---|---|---|---|---|
| 2022 | 43,077 | 21,371 | 20,858 | 15,095 | 8,192 |
| 2023 | 37,757 | 18,537 | 18,116 | 13,895 | 7,923 |
| 2024 | 30,486 | 14,599 | 14,532 | 12,340 | 7,304 |
| 2025 | 30,025 | 16,041 | 15,730 | 13,071 | 7,280 |
| Total | 141,345 | 70,548 | 69,236 | 54,401 | 30,699 |

[a]Same reported grade in both rounds. Another 595 records returned at a different grade.

had no register link. Among the 101,209 linked pupils who finished the regular wait list without an offer and whose April school was outside their eligible original report, the complementary offers in Table 5 account for 30,699 (30.33%). Search often expanded even when a former option remained on the public vacancy list: 45,848 of 46,940 such same-grade returners added a school. New supply, new information and an unsuccessful first round can all change a family's decision. The linkage establishes when an earlier omission preceded later enrollment. It cannot identify which family would pursue a particular omitted school under a hypothetical five-choice recommendation.

Those observed choices motivate the two tests in Section 9. One restores courses from the original report that the same pupil later named again. The other uses the newly named schools and the complementary-round vacancy file.

## 5 Why can a pupil lose a course that stayed on their list?

Consider a pupil placed under $A$ who lists the same course under $B$ and goes unplaced. Someone with higher priority must now hold that seat. Following this change leads to an applicant who would choose differently under the two lists.

Take two policies $A$ and $B$. They may change which programs appear, their order, or both. Programs keep their capacities and strict priorities. For each applicant let $\succ_i^A, \succ_i^B$ be the orders they would use to apply. In each order, place the outside option $\perp$ below all listed programs and above every omitted one. This convention records whether a program is available to an applicant. We still evaluate a first choice by the family's original report. For arbitrary stable assignments $M_A, M_B$ on the respective lists, define

$$W = \{h : M_A(h) \succ_h^A M_B(h) \text{ and } M_B(h) \succ_h^B M_A(h)\}.$$

Thus a witness ranks the assignment from each policy above the alternative when using that policy's list. No comparison between two omitted programs is needed.

Theorem 5.1. *Each applicant placed only under $A$ can be linked to a distinct person in $W$. If that applicant's former program remains on their $B$ list, the witness is someone else. The same claims hold separately for applicants placed only under $B$. Both bounds can be attained simultaneously.*

Proof. At each program cancel its common holders, then pair departing holders one-to-one with incoming holders until one side is exhausted. Start from an applicant placed only under $A$. At the current applicant $i$, the old assignment $j = M_A(i)$ ranks above $M_B(i)$ under $A$. If $M_B(i)$ ranks above $j$ under $B$, stop at $i \in W$. At the start this happens exactly when $B$ has omitted the lost program.

Otherwise $i$ wants $j$ under $B$. Program $j$ is real and listed under $B$, so stability makes it full of holders who outrank $i$. Its incoming holders cover all its departures. Follow $i$'s pairing to an arrival

$h$. Under $A$, $h$ ranks $M_A(h)$ above $j = M_B(h)$: either $j$ was omitted, or $h$ would otherwise block with $j$, where $h$ outranks the former holder $i$. This is the same condition we started with. Continue at $h$. Each applicant has at most one predecessor and the starting applicants have none. The paths cannot meet or cycle, so each ends at its own witness. Exchange $A$ and $B$ for the gain paths.

For equality take two one-seat programs $x, y$ and three applicants. One lists only $x$, one lists only $y$, and one has highest priority at both. Switching the last applicant's order from $x, y$ to $y, x$ displaces one specialist and places the other. The same person witnesses both changes. Disjoint copies attain any integer bound. □

*Binding shortlists.* When the two policies preserve the underlying order wherever both list a program, a witness must have an assigned program omitted on the other list. In the fixed-priority pooled replay, of the 2,527 Chilean applicants who lost their placement under the rank-informed rule, 693 no longer listed their old program. The remaining 1,834 still listed it, including 1,752 whose *entire* menu stayed the same. We traced these 1,834 losses to 1,834 distinct other applicants with edited menus. Reversing the comparison, 665 of 740 gains reached another person. In the ordinary and enriched NYC simulations of Section 8, 299 of 945 and 439 of 2,110 losses, respectively, occurred with the old apartment still on the list. Paths crossed up to seven programs in Chile and three in NYC. The trace certifies a chain in the two allocations. Several valid pairings can exist.

*Identical option sets.* With a common graph, a witness must be placed under both policies and reverse the order of their two assigned programs. If the orders also coincide, $W$ is empty, recovering invariance of the matched set [Roth, 1986]. When families try the recommendations first and then append every omitted eligible option (Section 9), 9,027 records met this reversal condition. We traced 1,871 losses and 540 gains to distinct witnesses within each direction. We also traced all 1,231 gross changes in the corresponding rental scenarios. As a finite check of the general result, we enumerated all pairs of stable outcomes across all ordered lists, strict priorities and capacity vectors in $\{0, 1, 2, 3\}^2$ in markets with three applicants and two programs. Both path constructions passed on 9,068,610 comparisons, including 8,867,592 with unequal graphs.

This question differs from asking which contracts are possible given only acceptability and one side's rankings [Haeringer and Iehlé, 2019], whether a single preference profile rationalizes observed assignments [Hu et al., 2020], or which new stable assignment lies close to an earlier one [Bredereck et al., 2020]. Here the two policies and assignments are given. The paths identify a distinct edited comparison behind each loss, including a loss experienced by an applicant who kept their former option.

## 6 How much of a menu change can a platform safely make?

Consider a district that has the submitted rankings and two proposed five-course menus. A family already placed under the first policy has a reasonable question about switching to the second: can I keep my place, or get one I prefer? The answer depends on when the district acts.

*Before the priority draw.* Start with fixed program priorities, as in the pooled-seat benchmark. For each possible draw $s$, clear the old menus to obtain $M^s$. To keep this allocation stable, each pupil must still see their assigned course. A newly shown course must also pass a test: if the pupil prefers it to their assignment, that course must be full with higher-priority holders. For an explicit finite set of draws, protect the union of required courses for each pupil and exclude the union of pairs that fail the test. Fill the remaining slots by score, separately for each pupil. The old menus ensure feasibility. This computes the best additive score under the requirement. In each protected draw, applicant-proposing DA on the new menus gives every pupil an assignment at least as good as $M^s$.

Allowing a fixed fraction of the listed draws to fail changes the optimization. An edit's cost now depends on which draws the other edits already cause to fail. The decision problem is NP-complete even with unit seats, two slots per editable pupil, and one optional substitution per row. The proof encodes graph edges as edits and their endpoints as the two draws in which a pupil loses a seat. A budget of $b$ failed draws can support $\binom{b}{2}$ edits exactly when the graph contains a $b$-clique. This hardness concerns an explicitly supplied distribution of orders. It does not establish hardness for the particular independent lottery simulator used below. Appendix D.2 gives the proof and the distinction between a fixed and a draw-dependent baseline.

The rank-informed menus gained 3,590 historical April-school hits over the rank-blind menus in 2022–24. We retained their common courses, froze each pupil's quota, and asked how many extra hits a designer with access to the April labels could gain by swapping the remaining courses. For the pooled benchmark, protecting 1,024 design draws per cohort permits 29,018 of 67,909 proposed substitutions. Even the hindsight optimum gains only 278 hits. A different menu that maximizes the number of substitutions loses 27 hits. Including more of the model's choices and including more pupils' eventual schools are distinct objectives (Appendix E).

Under Chile's reconstructed rules the comparable ceiling is **314**, or 8.75% of 3,590. This counts 96, 111 and 107 additional pupils in the three cohorts. We first bound what is possible by requiring the old detailed assignment to be feasible and stable in each grade, given the older pupils' baseline assignments. We then construct one menu profile per cohort that attains its bound. Re-clearing from the highest grade down returns the same course and seat type to every pupil in every one of the 3,072 cohort–draws. Thus 314 is the exact optimum for preserving the actual simulated assignments on these draws. The lottery comparison preserves the old within-tier draws and supplies the cross-tier comparisons required by typed seats. Only 252 of the newly coverable pupils appear in both models. A family lottery within courses gives 314; sharing family tickets across courses at a school gives 317. With the recorded priorities alone the same calculation permits 1,195 hits. The distinction is whether the designer must protect the assignment before learning which priority order will apply. Appendix F gives the certificate and checks what happens when the mechanism actually clears the edited lists.

*After tentative offers.* Suppose Ana and Ben see only $x$, which gives Ana priority. Ben goes unplaced. Ana prefers the vacant $y$ to $x$. Adding $y$ for Ana can place Ben without changing Ben's menu. At a school with several possible claimants, a move also has to respect people who already saw that school. For each program record the highest priority among applicants who saw it and preferred it to their old offer. A new holder must meet this threshold. Starting at the old matching, find a maximum flow of such moves from unplaced applicants to vacancies. Every moving incumbent gets a course they prefer, and every chain adds one recipient. Implementing any collection of complete chains requires at most one menu substitution per participant. DA on those menus preserves the witness's recipients and weakly improves their assignments. Dropping the program-priority test gives a second flow, an upper bound on what any reassignment can achieve while protecting applicants' welfare (Appendix J.2).

In Chile even the ceiling leaves $1,787 - 649 = 1,138$ fewer pupils placed than under the unchanged rank-blind menus. Requiring newly shown courses to be in pupils' original top five reduces the ceiling to four. In NYC, the chains recover the unchanged uniform headcount in 226 of 240 ordinary markets and every enriched market. The identity of recipients can differ across starting policies. Applying the same chain algorithm to both rental policies leaves uniform ahead by 128 and 51 placements respectively. The contrast concerns available acceptable moves in these markets, conditional on submitted school ranks and modeled rental preferences. It gives no estimate of leases signed or enrollment caused by a menu.

Table 6. How many additional people can be placed after the rank-informed or Haiku allocation? Direct referrals retain all incumbent contracts. Chains permit preferred moves. The ceiling relaxes program priorities. Deficit is the initial shortfall relative to the unchanged rank-blind or uniform policy. NYC counts pool repeated simulated renters.

| Starting point | Deficit | Direct referrals | Chains | Ceiling |
|---|---|---|---|---|
| Chile, pooled seats | 1,787 | 409 | 475 | 649 |
| NYC ordinary, 240 markets | 470 | 683 | 983 | 1,353 |
| NYC enriched, 240 markets | 898 | 1,926 | 2,493 | 2,844 |

## 7 The same five-choice rule in both markets

The two fitted recommenders use different inputs. A common benchmark lets us ask the same question in both settings. Suppose each person can list five options. What happens if everyone lists their five favorites? The alternative is to sample up to five feasible options uniformly without replacement for each applicant. In either case applicants propose in their original order. We keep any protected school entitlement outside the five slots. An applicant with at most five discretionary options sees them all. The benchmark uses modeled preferences in housing and families' submitted rankings in school choice. It tests what concentrating choices near the top of people's rankings can do at fixed capacity. The rental language model does not observe those realized rankings.

Table 7. First five minus uniform five. Both outcomes use all applicants as the denominator. Changes are percentage points. NYC rates use the study's geographic weights. School rates give each application record equal weight.

| Panel | Applicant instances | Full-list partner shown | Placed |
|---|---|---|---|
| NYC ordinary | 15,360 | +21.03 | −13.86 |
| NYC enriched | 15,360 | +17.03 | −18.88 |
| Chile 2022 | 570,891 | +0.82 | −0.23 |
| Chile 2023 | 536,353 | +0.99 | −0.23 |

Choosing favorites raised full-list partner coverage and lowered placement in all four panels. To compute expected rental coverage under sampling, each full-list recipient with degree $d$ contributes $\min(5, d)/d$. Placement requires running DA. We averaged 48 uniform draws per market; Monte Carlo standard errors on the estimated placement rates were at most 0.022 percentage points in NYC and 0.0012 in Chile. These standard errors describe the numerical uncertainty in averaging menus. They do not measure uncertainty about the housing behavioral assumptions or about future cohorts. Short lists and continuity protections in Chile mean that many pupils receive exactly the same choices under either rule. The larger rental differences do not show that a housing recommender is intrinsically more harmful. The shared direction shows that the reversal can arise under a common rule in both settings. The school and rental policy comparisons use the same placement measure.

*When keeping a favorite protects its recipient.* There is a precise reason to distinguish the school entitlement from an ordinary five-choice cap. If every list is a prefix of the applicant's full order, an applicant whose full-list DA partner survives that cut is placed at that program or one they prefer. To see this, run applicant-proposing DA on the prefixes, pausing anyone who exhausts their prefix. Continue that same run with the omitted suffixes restored. Those already holding an offer can only keep it or propose farther down their lists. Someone unplaced at the pause has already had every

prefix option rejected. Order independence of DA gives the full-list allocation at the end [Gale and Shapley, 1962, Gale and Sotomayor, 1985]. This is consistent with fewer placements relative to random menus: the guarantee concerns people whose full-list partner the prefix retains.

In the two rental designs every one of the 4,531 and 4,274 covered full-list recipients was indeed placed under first-five menus. In Chile an entitlement kept below the first five can break the prefix property. If an applicant ends up with a worse program while their full-list partner is hidden, their assignment must be one of those retained entitlements: a displayed top-five option could not be worse than an omitted one. The distinct witnesses from Theorem A.1 must therefore end at such a returning pupil. We verified this for all 126 covered-but-unplaced applicants in 2022 and all 120 in 2023. The school institution supplies a concrete boundary of the guarantee.

*Changing the rental ads and households.* We repeated the shared first-five comparison using separate survey donors and catalogues, without querying the language model. Among 8,400 supported markets, first-five increased partner coverage and reduced placement in both inventory designs in both evaluation windows. In September–October 2026 the ordinary-market changes were +20.66 and −15.26 percentage points, and the enriched-market changes were +17.62 and −20.04. In June–August 2025 the respective pairs were (+20.05, −15.30) and (+18.23, −18.92). The later enriched comparison covers only 79.97% of the intended borough–month weight: Staten Island lacked supply and September Manhattan had only 29 eligible low-price ads for 32 required draws. The other three panels cover 97.95%. Rates are normalized to the supported cells within each panel. These conditional simulations test the shared benchmark on different inventory and households. The Haiku comparison remains a development result.

## 8 New York City rentals

We paired archived model recommendations with a simulation of who could rent each apartment. The inputs are recent-mover household records from the 2020–24 American Community Survey and rental advertisements first listed in June–August 2026 [FirstMover, 2026, U.S. Census Bureau, 2026]. Each market contains 64 renter instances and 64 ads first appearing in one borough and month, treated as simultaneously available. An apartment must meet bedroom needs and cost at most 40% of monthly income. Lower-income households earn at most the survey-weighted median for NYC cash renters. Renter preferences combine rent, size, physical features, distance and taste shocks. Listing priorities depend on income relative to rent and a random shock specific to the renter–listing pair. We rerun applicant-proposing DA with exactly the same households, apartments and preferences under each display. The resulting placements are simulated. The ads provide asking rents, and the survey describes renter households. Neither source observes offers or leases between these parties.

In the ordinary design we sample households using survey weights and draw ads uniformly. In the enriched design half the inventory comes from a borough-specific affordability band. We then sample 32 lower-income households with a feasible option in that inventory and 32 higher-income households. This deliberate change lets us examine competition where lower-income households can participate. Each design has 20 seed blocks across three months and four supported boroughs, giving 240 paired markets and 15,360 renter instances. Staten Island had too few listings for draws of 64. The supported boroughs represent 97.95% of the intended recent-mover geographic weight.

The income cutoff is $64,435 in 2024 dollars. In asking-rent dollars at the study's July 2026 price base, the highest monthly rent allowed at the cutoff is $2,286. We use the calibrated preference specification from the rental pilot. It selects a rent coefficient of 4 and an idiosyncratic shock scale of 0.5 from a nine-point grid. Its burden calibration has an observable limitation. At 217 lower-income placements across the 240 ordinary calibration markets, even a feasible assignment

Table 8. NYC. Change under Haiku relative to uniform five-listing menus, in percentage points, with paired 95% simulation intervals. Both columns use every renter in the row. A renter without a full-list DA partner contributes zero to partner coverage in both policies.

| Design | Renters | Full-list partner shown | Placed by DA |
|---|---|---|---|
| Ordinary | All | +5.62 [4.96, 6.27] | −2.71 [−3.19, −2.23] |
| | Lower income | +0.11 [0.00, 0.24] | +0.00 [−0.19, 0.21] |
| | Higher income | +9.17 [7.99, 10.34] | −4.58 [−5.27, −3.86] |
| Enriched | All | +5.85 [5.04, 6.60] | −5.40 [−6.05, −4.73] |
| | Lower income | +0.78 [0.19, 1.43] | −0.92 [−2.23, 0.37] |
| | Higher income | +10.92 [9.47, 12.29] | −9.88 [−10.43, −9.36] |

chosen to minimize their first-quartile rent burden cannot push that quartile below 27.78%. The target quartile from recent movers' contract rents was 22.00%. These finite-market bounds concern the calibration exercise. The household survey does not validate the assumed listing priorities or individual rankings.

*What the recommender sees.* For each renter, the five-slot uniform rule draws from the feasible listings without replacement. The archived Claude Haiku 5.5 recommender receives the same candidate set, a budget, bedroom needs, target area, prices, physical features and distances. The prompt asks it to choose five listings. It has no observations of other renters' menus or the realized preferences. Both policies show all feasible options to anyone with five or fewer. The ads do not establish how long a unit remained on the market. We used the 20,740 saved valid responses from the development pilot. Households and apartments can recur across draws. Rates below pool individual indicators within each borough-month and then average cells with equal month weights and Census recent-mover borough weights. The pointwise intervals resample entire paired seed blocks 10,000 times. They condition on the catalogues, behavioral model and stored replies.

*What changed when everyone used the model.* Under the original pilot design the primary endpoint was the lower-minus-higher difference in hidden blocking pairs. The comparison here follows the same people and uses the saved menus. Ordinary markets placed 53.89% of renters under uniform lists and 51.18% under Haiku, even as full-list partner coverage rose from 9.42% to 15.04% (Table 8). The direction depends on the comparator. Against a fixed score that ranks eligible apartments using their observable features, Haiku placed 20.26 points more renters in ordinary markets and 24.47 points more in enriched markets. Its coverage differences were −0.44 and +0.66 points, respectively.

*What the group averages conceal.* In the ordinary design, 5,425 of 6,617 lower-income renter instances had no feasible listing. The standardized change in their placement rate was smaller than one hundredth of a point. In enriched markets the change was −0.92 points with an interval spanning zero, yet 1,203 lower-income instances lost a placement and 1,044 gained one. The enriched lower-income share with a hidden blocking pair (a renter and an unshown apartment that each prefer the other to their assignment) rose from 37.03% to 41.31%. The higher-income share fell from 85.19% to 78.01%. The absolute gap narrowed even as hidden blocking pairs became more common for the lower-income group. Partner coverage and placement were secondary outcomes in this development pilot. We treat these results as evidence of a mechanism under specified assumptions.

*Sensitivity to the rental preference model.* We kept the same households, inventories, feasibility rules, and saved recommendations, then changed one assumption at a time. We used income-blind listing lotteries, a common priority by income, a smaller rent penalty (1 instead of 4), and larger

idiosyncratic taste shocks (scale 1.25 instead of 0.5). Each scenario recomputes full-list partners and assignments. In all four, Haiku improved aggregate coverage and lowered aggregate placement in both supply designs. The lower-income result was more sensitive. In enriched markets the estimated placement change was −0.92 points under the calibrated model, +2.19 [1.13, 3.22] under common income priority, and −2.73 [−3.82, −1.61] under income-blind priorities. The ads and survey do not tell us which priority specification best describes real landlords. This sign change limits the distributional claim the simulations can support.

## 9 What if families look beyond the shortlist?

The platform shows five discretionary courses. Families can ignore it, restore an omitted choice, or look for a different school. We can observe what families sought after the actual regular round and how a risk warning changed applications in an earlier field study. We use each source for the question it can answer.

*Adding a school before the regular round clears.* Near a 30% predicted nonplacement threshold, a Chilean warning raised the probability of adding a school from about 19.9% to 41.5%. Most of the increase involved appending a school [Arteaga et al., 2022a,b]. We use each probability as a scenario for pupils who have an undisplayed school on their submitted list. In each scenario the same pupils are willing to respond under both menus. We select their highest-ranked course at a school entirely absent from that menu and run the complete institutional allocation again. In Table 9, coverage now credits every school pursued. The original displays still differ by 4,638 hits throughout.

Table 9. What happens when families pursue more than the five recommendations? Informed minus blind, among the same 1,969,120 register-linked pupils. An April hit means the historically attended school is in the pursued set. Fractional counts are means over eight paired repetitions; other rows are deterministic.

| How families pursue schools | April hits | Linked placements |
|---|---|---|
| Use the displayed menu | +4, 638 | −1, 670 |
| Append first omitted school, 19.9% participation | +3, 841.5 | −1, 664.3 |
| Append first omitted school, 41.5% participation | +2, 971.4 | −1, 656.0 |
| Restore that school in submitted order, 41.5% | +2, 971.4 | −1, 340.8 |
| Append courses the same pupil explicitly listed again | +4, 733 | −1, 653 |
| Append all omitted courses | 0 | −1, 234 |
| Restore all eligible courses in submitted order | 0 | 0 |

*Who chooses to keep looking?* Giving everyone the same response probability can conceal a reversal. Let the same families respond under both policies, and fix their number at 19.9% of pupils able to add a school under either. If they append their first omitted school, the sharp range of the April-hit difference is [−179, 5,412]. At the lower endpoint the rank-informed rule both misses 179 more eventual schools and places 1,669 fewer linked pupils. If instead the added course has the lowest demand relative to vacancies among omitted schools, even the least favorable common set of responders leaves a gain of 920 hits. The placement gap for that set is −1,612. The arithmetic behind the bounds is simple: a respondent matters for the comparison only when adding the course turns a miss into a hit under exactly one policy. The placement counts require national re-clearing. Appendix G gives the proof, the constructed responder sets and the sensitivity to unequal uptake across policies.

*Where families actually went.* Among 154,988 same-grade reapplicants, 139,190 named a school absent from their entire regular report. Only 20,698 had an eligible original course omitted by either hypothetical menu; 1,380 explicitly listed such a course again. Of 10,296 with an omitted course showing a positive complementary vacancy, 1,368 reapplied to one. Appending those exact repeat choices gives 4,733 more hits and 1,653 fewer linked placements. A different check uses the identities of the 154,988 actual returners but lets each append the highest-ranked school missing from their menu. Only 20,501 have such a school under either policy. The comparison gives 4,572 more hits and 1,665 fewer linked placements. The historical record points to a larger margin: families went looking for schools they had never listed.

To test that margin, we keep every simulated first-round place and give unplaced pupils access to the courses at new schools they actually named in the complementary round. Both arms face the same published second-round vacancies. Ignoring priorities, we find the largest possible number of extra placements in each arm. Across all applicants the two menus can add 63,571 and 63,847 recipients respectively; this shrinks the first-stage gap from 1,834 to 1,558. Giving register-linked pupils first claim on these vacancies leaves 1,419 fewer linked placements under the informed rule and 4,752 more pursued sets containing the April school. Families might reapply differently if their initial offers changed. These numbers describe how far their *recorded* search could reach in a common inventory. They do not predict the offers of an actual second round.

In the rental simulations, appending every omitted feasible apartment brings the enriched-market placement difference to +0.28 percentage points (paired block interval $[-0.07, 0.65]$). That interval includes zero; the corresponding ordinary-market interval remains below zero. The full set of comparisons appears in Appendix G.

## 10 Implications and limits

Adding rank helped predict the school later attended by pupils who received an initial offer, including those who enrolled elsewhere. Coverage fell slightly for the group without an initial offer. Under binding shortlists, it also increases first choices and leaves fewer pupils placed. The first-choice advantage reverses when families restore their highest omitted course at its original rank. The placement difference persists across the field-motivated response scenarios, while the coverage ranking can reverse for particular sets of families who keep looking. The social assessment therefore depends on which outcomes the platform values and how families use its advice. The families whose menus changed tended to move up in the replay. Families with the same menu who moved at all tended to move down. Preserving a tentative assignment protects these applicants by construction. Allowing recipients to move along complete chains gives the platform more scope to fill empty seats. The Chilean bound in Section 6 shows how little scope is left if every newly shown course must also be a top-five choice.

The rental experiment addresses a different social concern. A renter who cannot afford a single listing has zero hidden blocking pairs and zero chance of placement under either policy. A group can consequently score well on a hidden-options measure while having almost no access to housing. We keep these households in the denominator, report their feasible sets, and examine how much discretion the platform has once tentative offers exist. Giving lower-income renters priority among equally large referral assignments had room to move only three additional placements toward them in the ordinary draws. A service that also changes affordability or expands search would face a different feasible graph.

*Limits of the evidence.* The housing data identify asking rents and surveyed household characteristics. Preferences, housing-provider priorities, budgets and assignments are modeled. The budget in the prompt is correlated with income, even though group membership is absent from the

prompt. The language-model result conditions on one model version and one saved response per queried renter. The independent catalog check evaluates only the first-five benchmark. The data contain no broker relationships or commissions, and the study does not identify steering. In Chile the rank-informed model already has access to families' ordered applications. Its evidence concerns curation of known choices. The historical replay verifies each pupil's assigned course, including those of every other applicant competing for it. The public result does not name the category of seat a pupil occupied. Different underlying rules could also agree on the original reports and diverge on new ones. Our menu comparisons implement the published seat and family rules, and our lottery calculations specify how priorities are drawn. Family take-up remains a separate question. No family received these menus. Changing what a platform displays could change search, reported preferences, application, or acceptance. The complementary round records actual list expansion after an unsuccessful first round. The regular-round replays hold the original applicant pool, family links and underlying reports fixed. Our second-stage comparison adds observed complementary choices and a common stock of vacancies. It leaves open the identities of schools that families would discover because of a different offer or different advice. The Chilean application years may include some of the same children. We treat them as finite cohorts and report exact counts. The shared first-five benchmark uses the same rule in both settings. The site-specific recommenders address different decisions. We do not pool their effect sizes.

*What a platform can audit.* Record the available options and capacities, the two complete sets of proposed lists, the reference offer where one exists, and the assignment induced by each set of lists. Those records distinguish a successful recommendation to its intended recipient, a lost place on a list that never changed, and a person without any feasible alternative. The last case points to a constraint on supply. The first two call for evaluating recommendation and allocation together.

*Reproducibility.* The accompanying code regenerates the two studies' aggregate tables and figures. In NYC we checked 2,880 published market-by-group endpoint rows against the regenerated menus. The legacy reply cache stores listing indices without a digest of the original prompt payload. Its consistency with this reconstruction can be checked, while the exact original payload cannot be authenticated independently. In Chile we recomputed the blind, rank-informed and full-list assignments in the original two cohorts and checked all 3,321,732 applicant–policy outcomes against the archived serial solver. The 2022–23 sign criterion was recorded locally before data acquisition. The extension to 2024–25 was committed to the research repository before downloading either release. The joint transitions, alternative baselines, rollout, referrals, linked enrollment and responses to advice are follow-ups. Separate implementations rebuilt the eligible school choices from ministry files and verified the enrollment scores. A batched DA solver reproduced 64 national allocations used in the response analyses (32,714,384 applicant–policy outcomes). The two additional cohorts yielded a further 2,812,215 person–policy checks. For the institutional replay we independently joined predicted courses to the released initial offers. A second, synchronous implementation agreed on courses and seat categories in 600 generated two-grade markets. All shared school outputs are aggregates.

*Use of AI.* The team used OpenAI Codex extensively for mathematical arguments, code, numerical checks and writing. Appendix L describes this work. The rental recommender was a separate system (Section 8).

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## A What does showing a full-list partner guarantee?

The full-list partner in this section is an assignment in the same model as the restricted replay. The Chilean historical offer provides a separate empirical target.

*The best prediction can send two people to one seat.* Three applicants share three one-seat programs $x, y, z$. The vector of first choices, in applicant order, takes the values $(x, y, z)$, $(x, z, y)$, $(y, x, z)$ and $(y, z, x)$ with probabilities $2/6, 2/6, 1/6, 1/6$. Each state has distinct first choices, so all three receive those choices under full-list DA whatever the program priorities. The platform knows the probabilities but cannot see the state. With one slot per applicant, the unique choices maximizing expected partner coverage are $(x, z, z)$. They cover $5/3$ partners in expectation and place two people. The lists $(x, y, z)$ cover $3/2$ partners in expectation and place all three. Thus filling the third seat can cost as little as $1/6$ of an expected correct recommendation in this example.

*What an omitted partner can explain.* Let $H$ be the applicants whose full-list assignment is a real program missing from $D$. Let $K_j$ and $T_j$ count members of $H$ assigned to $j$ under $M$ and $N_D$, respectively.

Theorem A.1. *Relative to the full market, gross losses and gross gains in placement on $D$ are each at most $|H| - \sum_j \min(K_j, T_j)$. Furthermore, every unmatched applicant whose full-list partner is displayed can be paired with a distinct member of $H$ who is placed on $D$ and strictly prefers their full-list partner. Both conclusions hold for arbitrary stable selections on the two graphs.*

Proof. Delete $H$ together with the seats its members occupy in each assignment. Any remaining blocking pair would have blocked the original assignment. Restrict the residual full-market allocation to the residual display: every surviving contract is visible there, and removing other edges cannot create a block. The two residual allocations are thus stable on the same pairs, using capacities $c - K$ and $c - T$, where $c$ is the original capacity vector. For fixed pairs, adding one seat retains every matched applicant and adds at most one: regard seats as unit copies with common order within each program, run seat-proposing DA, and admit the new proposer last. An applicant who holds an offer can only trade up. Matched-set invariance transfers this comparison to any stable selection [Crawford, 1991, Gale and Sotomayor, 1985, Roth, 1986]. Comparing the two capacity vectors through their coordinatewise minimum bounds losses among the remaining applicants by $\sum_j (T_j - K_j)_+$ and gains by $\sum_j (K_j - T_j)_+$. Exactly $|H| - \sum_j T_j$ of $H$ lose a place, and none gain. Adding gives the bounds.

For the individual claim, at each program cancel the holders common to both assignments. Pair departing holders with incoming holders one-to-one until one side is exhausted. Start with an applicant who is unmatched on $D$ despite seeing their full-list partner $j$. Program $j$ must be full in $N_D$, and every new holder there outranks this applicant. Follow the pairing at $j$ to one of its new holders $h$. Full-list stability implies that $h$ had an assignment strictly preferred to $j$, since $j$ previously held a lower-priority applicant. If $h$ still sees that former assignment, repeat. Each step starts with someone who strictly prefers their former assignment and each pairing is injective. On a continued step the program is full after restriction. Its intake of new holders is therefore at least its number of departing holders, so the departing applicant always has a successor. The starting applicant has no incoming link, so the paths cannot cycle or meet. Each path ends at a placed applicant whose former assignment is hidden. □

For two policies $A, B$, one entry in the empirical comparison counts applicants with $C_i(B) - C_i(A) = 1$ and $P_i(B) - P_i(A) = -1$. The theorem gives each such applicant a distinct other person on the final display who is assigned a worse program than their full-list partner, which is hidden under $B$. That other applicant can have the same list under $A$ and $B$. It can happen with only two

applicants and one slot each. Let $i$ prefer $x$ to $y$, let $h$ prefer $y$ to $x$, and let both programs give $h$ higher priority. Full-list partners are $x$ for $i$ and $y$ for $h$. Display $A$ offers $i$ only $y$ and $h$ only $x$. Both place. Change only $i$'s list to $x$. Now $i$ sees their full-list partner and loses the competition for it. Under one-slot lists every witness is this direct: a new holder at the focal program could not also see their own strictly preferred full-list partner. The traced links in a larger market describe the two assignments and need not follow the chronology of DA proposals.

Let $p_{ij}(X)$ be the probability that $j$ is the full-list partner of $i$, given the information $X$ used to choose the lists. If a policy also uses randomness independent of the full allocation conditional on $X$, condition on its seed as well. By linearity, the conditional expected number covered is $\sum_i \sum_{j \in D(i)} p_{ij}(X)$. Selecting the most probable partners separately for each person therefore minimizes expected omissions, the coarse upper bound $|H|$ above. A smaller upper bound can coexist with fewer actual placements. For comparison, McCauley et al. [2026, Theorem 4.7] identify which hospitals can become unmatched under one-sided prefix pruning. Here omissions can occur anywhere in an incomplete list, and a displaced person can still see their own full-list partner.

## B A first choice and a place are different outcomes

Was the lower placement count the price of better assignments for other applicants? To find out, we ranked each assignment on the person's original list, before either shortlist was made. A renter's order is modeled. For a pupil we use the rank the family entered, distinguishing the ministry's appended continuity option. An unplaced person stays in the denominator at every cutoff. Conditioning on being matched would omit the very people whose losses we want to measure.

Table 10. Change in assignments at or above each rank. NYC compares Haiku with uniform menus, in geographically weighted percentage points. Chile compares the rank-informed predictor with the rank-blind predictor, in pupils. The first three school columns count only family-entered choices; the last includes any assignment, including an administrative fallback.

| | First choice | Top three | Top five | Any placement |
|---|---|---|---|---|
| NYC ordinary (pp) | +2.00 | +4.22 | +4.99 | −2.71 |
| NYC enriched (pp) | +2.87 | +6.69 | +7.72 | −5.40 |
| Chile 2022 (pupils) | +385 | +70 | −184 | −468 |
| Chile 2023 | +515 | +116 | −181 | −517 |
| Chile 2024 | +747 | +14 | −303 | −481 |
| Chile 2025 | +571 | −38 | −254 | −321 |

In every panel the new policy produces more first choices and fewer placements. Neither policy therefore dominates the other at every rank cutoff, in the sense of Featherstone [2020]. This conclusion also holds if we include each administrative fallback at its final list position: the first-choice and any-placement differences are unchanged. In the ordinary and enriched NYC markets, the paired 95% simulation intervals for the first-choice gains are $[1.67, 2.33]$ and $[2.41, 3.32]$ percentage points. These are ranks on the modeled full order, not evidence that particular renters would have signed a lease.

There is precedent for this pattern under a different policy. In Chilean simulations Correa et al. [2022, Table 6] found that changing from course-specific lotteries to a common lottery increased both first choices and nonassignment. Mukhopadhyay et al. [2026] also compare tie rules using several welfare measures. Our counterfactual keeps the priority order and quota fixed, changes the predicted shortlist, and observes which affected families had the same list under both policies.

A simple accounting exercise puts a price on the disagreement without estimating anyone's utility. Give one point for any placement and an additional $\lambda$ points for a first choice. On the four school cohorts together, the change in this index is $-1{,}787 + 2{,}218\lambda$. Someone using this objective would prefer the rank-informed rule if and only if $\lambda > 1{,}787/2{,}218 \simeq 0.806$, with a tie at equality. Nothing in the data identifies that bonus or implies it is the same for all families. Counting assignments alone implicitly sets it to zero.

Who moved up? Ordering the outside option last and an appended continuity option after all family-entered choices, 8,076 Chilean applicants with edited lists got a strictly better result and 2,200 got a worse one. Among applicants with identical lists, 2,254 improved and 7,215 worsened. Each of the four years has this same split in direction. These are paired changes between two full-market replays. The two groups differ in list length and exposure to competition. Comparing their rates would not identify a treatment effect of editing one's own list.

## C Who loses a placement?

At the level of a whole market, both studies give instances of higher coverage and fewer placements. The losses reach different people. Figure 1 partitions everyone who was placed under *A* and unplaced under *B*. We first separate people whose menu was identical. Among the rest, we ask whether their full-list partner became visible, became hidden, or had the same visibility as before.

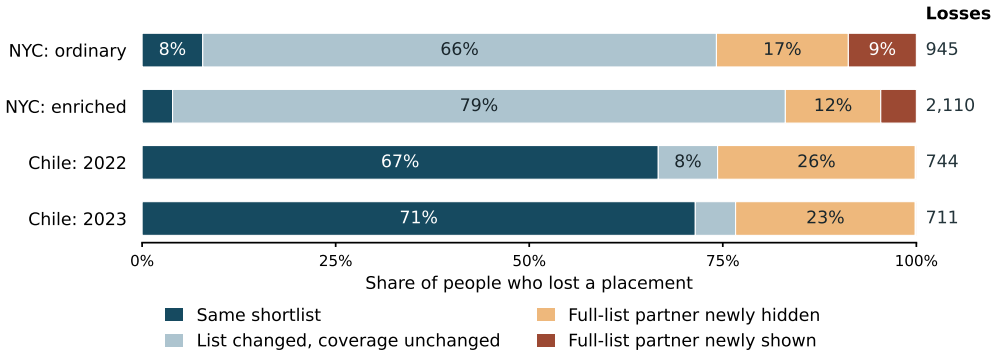



Fig. 1. What changed for people who lost a placement? Bars within each row sum to 100%. Numbers at right are raw gross losses. A changed list can retain the same partner coverage, including when the applicant had no full-list partner. The latter applicants still count in the analysis.

*A change that reaches someone else's list.* In the original 2022–23 Chilean cohorts, only 32,914 of 1,107,244 application records had different displayed courses under the two predictors. Of 1,455 applicants who lost a placement, 1,004 had exactly the same personal menu. The reverse happened to 406 of the 470 who gained one. Holding preferences, priorities, capacities and an applicant's own list fixed, a change in that applicant's placement can arise only through competitors' menus. These spillovers account for a net loss of 598 seats among applicants whose lists stayed fixed. Among the 32,914 applicants with edited lists, 451 lost a place (1.37%). Among the 1,074,330 with identical lists, the 1,004 losses represent 0.093%.

In the ordinary NYC markets, 945 renter instances lost a placement and 475 gained one. Of the 945 who lost, 74 saw identical menus and 83 newly saw their full-list partner. In enriched markets, 2,110 lost a placement and 1,212 gained one. Here 83 of those who lost saw identical menus and 97

newly saw their partner. Over both designs, newly seeing one's full-list partner and losing a place happened to 180 renter instances. For all 180 we found the distinct displaced-partner witnesses in Theorem A.1 within their respective market. Of these paths 169 reached a witness after one program. Longer paths involved up to three programs.

*The newly shown school offer often converted.* In Chile in 2022–23, 3,750 applicants newly saw their full-list DA partner under the rank-informed predictor. Of these, 3,728 obtained that exact course in the replay and 20 obtained a different course. The remaining two went unplaced. In ordinary NYC markets 1,056 of the 1,720 newly shown renters obtained the named apartment (61.4%); in enriched markets it was 1,277 of 2,012 (63.5%). Across all applicants in each NYC design the number receiving their exact full-list partner increased by 523 and 553, respectively. Across the two school cohorts it increased by 5,049. More people therefore received their exact full-list partner even while fewer received any placement. We verified the individual witness path for each of the two newly covered school applicants who went unplaced. Analyzing the recorded historical offer as a separate target, 3,366 of the 3,622 applicants who newly saw that program received it in the replay. Thus a platform could document a high rate of placement at the newly shown program and still miss a loss elsewhere in the market. The witness theorem applies to the modeled full-list partner. It uses stability and cannot be applied to an arbitrary historical offer.

*Feasibility and priorities.* Let $\nu(D)$ be the largest number who can simultaneously take displayed options within capacity, and $g(D) = \nu(D) - |N_D|$ the feasible places unrealized by DA. The placement change is $\Delta\nu - \Delta g$ [Aguiar and Hong, 2026]. In ordinary NYC markets, displayed capacity fell by 3.84 percentage points and the shortfall shrank by 1.13, giving the observed 2.71-point decline. In enriched markets those changes were $-12.29$ and $-6.89$ points. The school results differ. In 2022 the display could accommodate 244 fewer applicants and the shortfall grew by 224, for a loss of 468. In 2023 those figures were 153 and 364, for a loss of 517. Each school maximum is certified by a feasible flow and a cut of equal value. This benchmark uses only displayed edges and seat counts. It relaxes priorities and the obligation to place a particular returning pupil. These identities describe two margins on the same graph. They do not isolate the causal effect of changing priorities.

## D Changing lists while protecting tentative offers

Suppose the platform observes an allocation $M_A$ stable on the old lists before deciding who receives new lists. Both menus retain the same underlying ranking over programs. Each person can receive either their entire old row $A_i$ or their entire proposed row $B_i$. Flag row $i$ if either (i) $M_A(i)$ is a real program missing from $B_i$, or (ii) $B_i$ contains a pair $(i, j)$ that would block $M_A$. For the second test, $i$ must prefer $j$ to $M_A(i)$, and $j$ must have a vacancy or prefer $i$ to one of its holders. Since $M_A$ is stable on $A$, only newly added options need this test. Let $F$ be the flagged applicants. With indexed rankings, check feasibility of each old contract and compute each program's occupancy and worst incumbent priority once. A scan of the proposed edges then constructs $F$ in linear time in the explicit menu and capacity input size. This rule uses baseline assignments and the rankings relevant to these pairs. In a simulation it can use the modeled rankings; a service with incomplete information would need to elicit these comparisons.

Theorem D.1. *If $S$ receives its proposed rows, $M_A$ remains stable exactly when $S \cap F = \varnothing$. On any such display every stable outcome has the original recipients, and applicant-proposing DA weakly improves every applicant's assignment relative to $M_A$. For an additive row objective with gain $g_i$ from switching row $i$, the largest net gain that keeps $M_A$ stable is $\sum_{i \notin F} (g_i)_+$.*

*More generally, put $R = S \cap F$ and let $N$ be any stable outcome on the mixed display. Let $K_j$ and $T_j$ count people in $R$ assigned to program $j$ under $M_A$ and $N$, respectively. The gross numbers of placement*

*losses and gains are each at most*

$$\left|\{i \in R : M_A(i) \neq \perp \text{ or } N(i) \neq \perp\}\right| - \sum_j \min(K_j, T_j) \;\leq\; |R|.$$

Proof. A removed baseline partner makes $M_A$ infeasible. A newly displayed blocker prevents its stability. Both defects depend only on the applicant's row when the allocation being tested is fixed. In their absence every old contract survives and every displayed pair passes the blocking test. Matched-set invariance and applicant optimality give the first guarantees. The objective is therefore maximized by switching precisely the unflagged rows with positive gain, with free choice on zero gains.

For the bound, remove $R$ and its occupied seats from both allocations. In $M_A$ all remaining assignments survive on the mixed display and no remaining pair blocks. In $N$ stability also survives the restriction. Compare the resulting capacity vectors $c - K$ and $c - T$ using the one-seat argument of Theorem A.1. Outside $R$, losses are at most $\sum_j (T_j - K_j)_+$ and gains are at most $\sum_j (K_j - T_j)_+$. Within $R$, losses equal $\sum_j K_j - d$ and gains equal $\sum_j T_j - d$, where $d$ counts members matched under both allocations. Adding either margin yields the stated expression. □

The condition protects the old allocation as a stable witness. A flagged applicant can still switch without causing any placement loss. The distinction allows the rule to be checked before reallocation and limits the gains it can deliver. For independent switches with probabilities $p_i$, the probability that the old allocation remains stable is $\prod_{i \in F}(1 - p_i)$, and expected gross losses or gains are at most $\sum_{i \in F} p_i$. The probability factorization is related to the one-sided uncertainty result of Aziz et al. [2020]. Here program priorities stay fixed and applicant menus change.

For a fixed reference partner, $g_i = C_i(B) - C_i(A)$. Maximizing this objective using the partner's identity gives a hindsight ceiling. In the 2022–23 school cohorts, 27,851 of 32,914 changed rows could switch together while supporting every old contract. Only 22 of the 3,750 newly shown full-list partners occurred on those rows. A platform could reasonably object: why choose between two complete lists when it can replace just one option? We address that objection next. Appendix I separately studies random partial adoption without protection.

### D.1 How much does changing one slot at a time buy?

Fix the same baseline assignment $M_A$. For applicant $i$, let $U_i$ be a candidate pool containing $A_i$, and let $q_i = |A_i|$ include any protected slots. The required set $L_i$ contains the applicant's baseline partner if matched and all policy-protected options (in Chile, the continuity course if entitled). Let $V_i$ consist of the options in $U_i$ that do not form a blocking pair with $i$ at $M_A$. Inspecting a pair requires the applicant's preference relative to the current assignment and the program's vacancy or priority cutoff at that assignment.

Proposition D.2. *The menus with exactly $q_i$ options per applicant that respect entitlements and keep $M_A$ stable are precisely*

$$L_i \subseteq D_i \subseteq V_i, \qquad |D_i| = q_i \quad \text{for every } i.$$

*For any additive edge score, an optimal such menu keeps $L_i$ and fills the remaining slots with the highest-scored options in $V_i \setminus L_i$. For a single known reference option per applicant, the exact coverage gain over baseline is the number of previously hidden reference options in $V_i$ whose applicant has a free slot $q_i > |L_i|$.*

Proof. Removing a required partner makes $M_A$ infeasible; revealing any blocker makes it unstable. Conversely, if the required edges survive and no blocker is visible, $M_A$ is stable on the new display. This gives exactly the displayed family of sets. It is nonempty since the old list is

feasible in every row. Exchanging a lower-scored optional edge for a higher-scored eligible one can only increase the objective. For the coverage claim, give each reference edge score one and break ties in favor of old options. Every old hit can be kept, and each admissible missing reference with room in its own row can be added independently. □

The blocking-pair test has a direct predecessor: Boehmer et al. [2021, Observation 1] show that deleting blocking edges suffices to stabilize a specified matching in one-to-one markets. Here the quantity of interest is how much of a proposed recommender's objective can be retained at a fixed quota, with institutional entitlements. As in Theorem D.1, the original contracts form a stable witness. DA may select a different allocation, but every original recipient remains placed at a weakly preferred option and no previously unplaced applicant enters. The rule requires a known tentative assignment and the comparisons just described. It cannot be run from scores alone.

Table 11. Additional full-list partners shown over the original menus. "Whole" and "Slots" are hindsight maxima under the stable-witness requirement. "Score" edits individual slots over all feasible options without seeing the target. "Switch" adopts the proposed policy without the requirement. Counts pool applicant instances within each setting.

| | Whole lists | Slots, $A \cup B$ | Slots, all | Score | Switch |
|---|---|---|---|---|---|
| NYC ordinary | 132 | 591 | 2,014 | +243 | +867 |
| NYC enriched | 133 | 662 | 3,018 | +193 | +912 |
| Chile 2022–25 | 44 | 51 | 74 | +10 | +5, 975 |

Why is the school ceiling so low? Of 1,885,296 full-list partners in the four cohorts, the blind menus already showed all but 9,407. Of those misses, 9,333 would themselves block the old assignment if displayed. No amount of rearranging the other four slots can make those pairs compatible with that assignment. Every one of the remaining 74 can be added; the quota rules out none. The conclusion changes in the rental markets. Of 6,377 missing partners in the ordinary markets, 2,014 could still be shown, as could 3,018 of 8,728 in enriched markets. This is a difference in the composition of the missed targets, in addition to the much higher starting coverage in Chile.

The two benchmarks use different information. A clearinghouse that knows all inputs and tie orders can compute its own full-list DA target. A predicted historical offer and an apartment allocation under unobserved tastes are different objects. For the score column we instead maximize the frozen rank-informed school score, or the number of the rental model's proposed picks retained (the archive records selections, not scores for every listing). Neither uses the target. Under these edits DA retained the same people in every market and gave 67 and 80 renters a strictly better modeled assignment across the two designs. Across the four school cohorts, 12 pupils improved. For the separate historical-offer target, the score rule actually *lost* 30 hits in Chile; the hindsight maximum was a gain of 645. The rule guarantees protection, not an improvement on an unknown prediction target. Nor is 74 an upper bound if contracts may move and the only requirement is retaining the same headcount.

*Improving prediction and placement together.* Yes in these cohorts, if it has more information. Consider a clearinghouse that already knows all reports, capacities and realized priority orders. Compute full-list DA, keep each matched applicant's full-list partner and any entitlement, then fill the other slots by the frozen rank-informed score. All full-list contracts remain feasible; restricting a stable display cannot create a blocking pair for that assignment. The full-list allocation is therefore stable on the new menus. The set of recipients stays the same as in full-list DA, although a person's

exact partner can change. This construction fits the five discretionary slots because each applicant needs to retain at most one partner in addition to the entitlement.

Compared with the unrestricted rank-informed predictor, this rule showed *more recorded historical offers* in every year (+856, +814, +660, +489) and also placed more pupils (+42, +60, +39, +38). It used no historical offer as input. This rules out an inherent prediction–placement frontier even for the observed target. The guarantee protects the full-list allocation, a different baseline from the rank-blind seats protected in Table 11. A service advising applicants before reports or tie orders exist cannot simply run this rule, and the housing ads do not supply the required preferences.

### D.2 Choosing a list before the lottery

Suppose a platform has $m$ possible priority draws, listed explicitly. A draw supplies a strict priority order at each program. Applicant preferences, candidate pools, capacities, scores and protected options are fixed. On draw $s$, let $M^s = \mathrm{DA}_s(A)$ be the assignment on the old menus. The platform must choose a single display $D$ before it learns $s$. Call a draw a failure of the stable-witness requirement if $M^s$ is infeasible or unstable on $D$. The old assignment can vary by draw.

Theorem D.3. *For an explicit list of equally likely draws, the maximum additive score subject to at most $b$ witness failures is computable in polynomial time for each fixed $b$. When $b$ is part of the input, deciding whether score $K$ can be reached is NP-complete and W[1]-hard parameterized by $b$. The hardness already holds with unit seats, a common priority order at all programs in each draw, the same baseline assignment in every draw, and two options on each editable menu. The same NP-completeness and parameterized hardness hold if a draw instead counts as a failure only when some old recipient loses their placement under DA.*

Proof. For any nonempty set $H$ of draws to protect, define $L_i^s$ and $V_i^s$ as in Proposition D.2, using $M^s$ and the priorities in $s$. The required options and the eligible options for row $i$ are respectively

$$L_i(H) = \bigcup_{s \in H} L_i^s, \qquad V_i(H) = \bigcap_{s \in H} V_i^s.$$

A single display protects all draws in $H$ exactly when $L_i(H) \subseteq D_i \subseteq V_i(H)$ and $|D_i| = q_i$ for every $i$. This follows by imposing Proposition D.2 in each draw. The old menus meet all these constraints. Retain the required options and take the highest-scored eligible options in the remaining slots. For $0 \le b < m$, enumerate the $\binom{m}{b}$ sets of draws to allow to fail and optimize on each complement. A display failing fewer than $b$ draws is considered since its failed draws can be extended to a set of size $b$. For $b = m$, fill the menus using only the institutional protections. This gives a polynomial algorithm for fixed $b$, including $b = 0$. The enumeration concerns the witness requirement. It does not establish a polynomial algorithm for a placement-only requirement on unrestricted markets.

For hardness, start with a simple graph $G = (V, E)$ and a clique target $2 \le k \le |V|$. Create one draw per vertex and two applicants per edge $e$. Applicant $i_e$ ranks $y_e \succ x_e \succ z_e$ and must show two of these options, including $x_e$. Their old menu is $\{x_e, z_e\}$. Applicant $h_e$ lists only $y_e$. All three programs have one seat and occur in no other edge's lists. The only positive score is one for displaying $y_e$ to $i_e$. On the old menus every $i_e$ gets $x_e$ and every $h_e$ gets $y_e$. In draw $v$, rank $i_e$ above $h_e$ exactly when $v \in e$. Concatenating these ordered pairs in a fixed edge order gives one strict priority order common to all programs in that draw.

Let $S$ be the edges whose applicants switch to $\{x_e, y_e\}$. If $v \in e \in S$, applicant $i_e$ takes $y_e$ from $h_e$, who then has no placement. If $e \notin S$ or $v \notin e$, both old placements survive in the gadget. Thus the failed draws for either requirement are exactly $\bigcup_{e \in S} e$, and the score is $|S|$. With $b = k$ and $K = \binom{k}{2}$, the score is attainable exactly when $G$ contains a $k$-clique: a simple graph on at most $k$ vertices has at most $\binom{k}{2}$ edges, with equality only for a clique. Both decision problems are in NP by guessing

the menus and checking each explicit draw, using DA for the placement criterion. The reduction preserves the parameter $k$, and parameterized Clique is W[1]-complete [Downey and Fellows, 1995, Corollary 3.2]. □

Hardness also applies at a fixed percentage of failures. Fix a rational $0 < \alpha < 1$. Before the reduction, add $p$ mutually adjacent vertices joined to every original vertex, choosing the smallest $p \geq 0$ with $k + p \geq \alpha(|V| + p)$. Then add isolated vertices until the graph has $m = \lceil (k + p)/\alpha \rceil$ vertices. These additions take polynomial size for fixed $\alpha$, preserve the clique decision at target $k + p$, and give $\lfloor \alpha m \rfloor = k + p$. If distinct priority profiles are required, append $m$ applicants with one private seat each. Use a different cyclic order of this irrelevant block in each draw. The distribution above is uniform on the listed scenarios. It makes no assumption of uniformity on the set of all priority permutations.

The reduction realizes a familiar combinatorial problem inside a matching market. Each edit earns a point, and distinct failed draws count against a shared budget even when several edits fail on the same draw. This is the set-union knapsack objective of Goldschmidt et al. [1994]; the construction makes these sets arise from actual lost placements. Related work optimizes the choice of a matching across uncertain preference profiles [Aziz et al., 2020, Chen et al., 2018]. Here the design variables are the displayed edges and the old assignment in each draw is given by the baseline policy. For one predetermined assignment and all refinements of tied priorities, stability under every refinement is the classical super-stability condition [Irving et al., 2000, Proposition 2]. Protecting the old DA assignment separately in each listed draw permits those assignments to differ.

*What changes under a uniform lottery?* The distribution of priority orders matters. Suppose the baseline assignment is the same $M$ on every draw in the support, program priorities have fixed tiers, and each pair within a tied tier has equal chances of coming first. This includes a uniform common lottery and independent uniform program lotteries. For any menu, either $M$ stays stable on every supported draw or failure has probability at least $1/2$. A dropped partner fails on every draw. Otherwise, take a pair that blocks on some draw. The applicant always prefers the program to their partner in $M$. A vacancy or an incumbent in a strictly lower tier gives a certain block. In the remaining case, the applicant outranks a particular tied incumbent with probability $1/2$, which suffices to block. Consequently every failure tolerance below one half gives the same witness-constrained optimum as zero tolerance in this special case. With two tied applicants contesting one incumbent's seat, the threshold is attained.

A draw-dependent baseline changes the conclusion even under a uniform common lottery. Give every program one seat. Applicant $a$ lists only $x$. Each of $t \geq 1$ others lists only $y$. Applicant $i$ changes from $x, z, w$ to $x, y, z$, ordered as written, with private $z, w$. The only possible blocking edge at the old outcome is $(i, y)$. It blocks exactly when $a$ precedes $i$ and $i$ precedes all $t$ original applicants to $y$. Then $i$ takes $y$ and its former holder goes unplaced. This event has probability $1/[(t + 2)(t + 1)]$. An edit can therefore violate the guarantee on an arbitrarily small share of draws even when every old partner stays available.

## E What a finite lottery certificate buys in Chile

This appendix reports the original pooled-seat benchmark. Appendix F gives the corresponding institutional results. The common options and entitlements on the two frozen menus stay fixed. Suppose there are $d_i$ old-only and $d_i$ proposed-only options for pupil $i$. Among the old-only options let $p_i$ count those that the pupil receives in any protected draw. Among the proposed-only options let $f_i$ count those blocking at least one protected allocation. At most $d_i - p_i$ old slots can be freed and at most $d_i - f_i$ new slots can be used. Any eligible new slot can replace any removable old one

for this pupil. Thus the exact number of substitutions toward the new policy is $d_i - \max(p_i, f_i)$. We use submitted rank to break ties. The objective counts proposed choices adopted. An additional choice need not be a useful prediction.

*Design.* For each of 2022–24 we drew 16 priority profiles using independent within-course permutations inside the five public priority categories. We solved the zero-, one- and two-exception problems exactly, enumerating the possible draws to exclude. We evaluated on the recorded within-course order and on 64 fresh draws shared by baseline and challengers. We fixed these choices before computing the new outcomes on these previously studied cohorts. With exception budgets of zero, one and two, the optimal menus retained 39,293, 39,727 and 40,189 proposed substitutions, respectively. Under the recorded orders 146, 158 and 184 former recipients lost their places. The entire switch made 67,909 substitutions and displaced 2,088. A simulation batch is a specification of possible priorities rather than an estimate of Chile's actual priority law. We exclude 2025 because the tie procedure changed.

*Training support and edit-matched comparisons.* After seeing the first test we extended the nested training support to 64, 256 and 1,024 draws, always requiring the old assignment to be stable on all training draws. For each resulting menu we made an alternative with exactly the same number of slot substitutions and changed rows. We permuted the numbers of swaps among pupils with the same number of differing slots, then sampled which old-only and new-only options to exchange. This control retains all shared courses and protected options. Eight seeds are evaluated at the recorded priorities, and the first seed also on every simulated test draw.

Table 12. Nested training supports on the same 64 test draws per year. Failure counts pool 192 cohort–lottery realizations. A certificate fails when a baseline pair is missing or has a blocking pair. A gross exit is a baseline recipient who loses their placement. Repeat appearances in different lotteries count separately.

| Design draws | Adopted slots | Failed certificates | Draws with exits | Gross exits | Control exits |
|---|---|---|---|---|---|
| 16 | 39,293 | 192 | 192 | 9,491 | 78,315 |
| 64 | 33,452 | 192 | 189 | 1,087 | 67,499 |
| 256 | 30,709 | 161 | 103 | 161 | 62,575 |
| 1,024 | 29,018 | 52 | 30 | 33 | 59,768 |

On the recorded orders, none of the three 1,024-draw menus cost an incumbent their placement. In two cohorts the old assignment nevertheless lost its stability certificate. The eight edit-matched comparison seeds produced 894–968 gross exits, summing three cohorts separately for each seed. Table 12 uses the same simulation holdout we had already inspected at 16 draws.

We then froze the 1,024-draw design and ran 256 additional draws per cohort with a fresh seed batch. Across these 768 cohort–lottery realizations the screen caused 87 gross exits and 91 entrants. There was at least one exit in 26, 34 and 23 draws by year, and at most two exits in any single draw. The certificate failed in 55, 76 and 56 draws. The edit-matched control using the first seed caused 237,115 gross exits across the same 768 realizations. At the recorded orders, the screened menus improved net placement by one, placed five baseline recipients at worse ranks and gave two fewer first choices. They showed 14 more initial offers and 27 fewer April schools than the blind predictor. The unrestricted switch showed 4,228 more initial offers and 3,590 more April schools. Each historical label counts an applicant at most once, including where more than one listed course maps to the same school.

*Could another protected menu recover the predictive gain?* For each historical target separately, imagine knowing every pupil's label. If the target already appears on $A_i$, keep that row. Otherwise one new hit is possible exactly when (i) some proposed-only option matches that target and passes every protected draw, and (ii) at least one old-only option is removable. These conditions use fixed assignments and hold independently by row, so summing the indicators gives the exact maximum increase over the blind menu. A course-level initial offer and a school-level April enrollment each get their own maximizing display. This is an information upper bound over the union of the two menus with shared choices protected.

With no lottery constraint, the union permitted up to 4,823 additional April-school hits. Requiring stability in all 16, 64, 256 and 1,024 design draws lowered that number to 593, 353, 302 and 278 respectively. For the last support, the maximum initial-offer gain was 48. Even the best feasible use of the historical labels could reproduce at most 278/3,590 = 7.7% of the unrestricted rule's net increase in April-school coverage. This limit concerns the stated menu pool and the requirement to preserve each old assignment as a stable outcome. Protecting only whether a pupil remains placed would give a different design problem.

## F Reconstructing the regular round and rerunning the ceiling

*Inputs and historical test.* The ministry's A1 file identifies courses and seats by category, B1 describes applicants, C1 records each ranked application, priority flags and realized lottery position, and F1 identifies family links. We constructed a market using those four files, initialized its historical-outcome field to missing, and saved the predicted course for every pupil. In a separate step we joined predictions to the initial school and course in D1 using the masked applicant identifier. Both an offer at another course of the same school and an incorrect no-offer prediction count as errors. April enrollment, family responses to offers and subsequent wait-list assignments are absent from this validation target.

We used the seat orders in Correa et al. [2022, Tables 1–2] and the reconstruction in Doğan et al. [2026, Appendix S7]. At a regular seat, priority descends from returning pupils to enrolled siblings, newly assigned higher-grade siblings, staff children, former pupils and other applicants. At disadvantaged seats, disadvantaged status follows the sibling tiers. At special-education and academic seats, returning pupils come first and eligible applicants follow. The released selection order applies where the program runs a separate selection procedure. A pupil can try more than one seat type at the same course. The order depends on the type of course and the pupil's return, sibling, disadvantage and special-education status. If returning pupils exhaust the published seats at their protected course, each can use a personal fallback there. It cannot be given to another applicant.

The highest grade clears first. A sibling already assigned to a school in a higher grade can change a younger applicant's priority there. For a family who requested block processing, the younger applicant tries displayed courses at the older member's school first, preserving the submitted order within the promoted group and within the remaining courses. These are separate updates. Each replay recomputes both from its own assignments. Same-grade links do not promote each other's schools. Holding the originally submitted reports only down to the guaranteed course leaves 141 discrepancies, even after these rules are in place. The underlying order can place a course below the guaranteed course and the family rule can promote it. Restoring those rows in the historical market accounts for the last 16, 39, 42 and 44 mismatches respectively. The two frozen menu policies still use their original candidate pool.

*Comparing menus before a priority draw.* The matched rerun uses the original 2019 scores, rank-blind and rank-informed display masks, quotas, applicant cohorts, labels and 1,024 scenario seeds. It

Table 13. Agreement with the initial course or no offer in the four national cohorts. Each row adds the stated change to the preceding row. The increments depend on this order.

| Allocation on original lists | Exact outcomes | Remaining discrepancies |
|---|---|---|
| Pooled seats and fixed priorities | 1,963,324 | 81,325 |
| Published vacancies and personal return | 1,979,498 | 65,151 |
| Separate seat categories and proposal orders | 1,981,888 | 62,761 |
| Priority from newly assigned older siblings | 2,002,303 | 42,346 |
| Opted-in family preference promotion | 2,044,508 | 141 |
| Restore submitted courses below the protected course | 2,044,649 | 0 |

preserves the relative lottery order within every original course and priority tier. Pooled priorities required no tie comparison between two different tiers. To obtain the raw order needed by the typed-seat rule, we interleave the ordered tier lists at each course uniformly, independently across courses. A uniform permutation of the multiset of tier labels implements that conditional draw. Each model computes its own baseline allocation. The two additional family lotteries draw a uniform order over the connected components of F1 and then over applicants within a component, either separately at each course or once per establishment. All other institutional rules stay fixed.

Table 14. Maximum additional historical April-school hits above the blind menus. Each design column protects the stated number of nested draws per cohort. The recorded column uses only that year's realized priorities. All models allow 4,823 hits with no allocation constraint; the unrestricted switch gains 3,590 net hits.

| Rules and priority generator | Recorded | 16 | 64 | 256 | 1,024 |
|---|---|---|---|---|---|
| Pooled, original course draws | 1,275 | 593 | 353 | 302 | 278 |
| Institutional, matched course draws | 1,195 | 623 | 394 | 336 | 314 |
| Institutional, family tickets by course | 1,195 | 614 | 393 | 338 | 314 |
| Institutional, family tickets by school | 1,195 | 601 | 380 | 337 | 317 |

At 1,024 matched draws the maximum number of adjustable slots a platform could switch was 30,520 of 67,909. The April target counts pupils, so the optimum for that target can be achieved with one edit per additional hit and all other menus unchanged. Of the 278 pupils reachable under the original benchmark, 252 remain reachable under the reconstructed mechanism; 26 leave and 62 enter. The same calculation for historical *initial offers*, a course-level target, admits 26 additional hits under the reconstructed rule and 48 under the pooled benchmark. Both label calculations give the designer information unavailable when a real menu is issued.

*Which lottery structures look plausible in the released data?* The Ministry reports a realized lottery position on each application. This lets us validate the historical allocation separately from assumptions about how to draw another lottery. The designers describe drawing an order over families, then drawing an order within each family at a school [Correa et al., 2022, Sec. 4.1.2]. Article 36 of Chile's admissions regulation specifies an ordering for an applicant to a particular course at a school [Ministerio de Educación de Chile, 2016]. We compared pairs of students who both applied to two courses at the same establishment. Their relative order reversed in 891,182 of 1,822,995 student-pair/course-pair comparisons (48.89%). A single strict order shared by every course at an establishment would give zero reversals. The 2022–24 data also show dependence within families. Among 53,107 course-level groups containing at least two members of one connected component

of the published family links and at least one outside applicant in the course, 51,117 groups (96.25%) occupied consecutive positions in the realized order. Independent uniform orders over individuals would give 2,833.49 contiguous groups in expectation (5.34%) at the same course and group sizes. These counts describe the three released orderings. They leave the dependence between unobserved draws undetermined. We use family lotteries as sensitivity analyses and validate historical offers with the recorded ordering.

*Computing the institutional ceiling.* A displayed course now permits several seat types. For a priority draw $s$ let $M^s$ be the detailed rank-blind allocation, including each assigned course and its seat type or continuity fallback. Given the older pupils' assignments in $M^s$, each younger pupil has a definite order of courses and seat types, and each seat type has a strict priority. Promoting listed courses at an older sibling's school preserves the relative order of all retained courses. For this fixed assignment, a change of menu therefore restricts the same ordered list of potential contracts in that grade.

To keep $M^s$ feasible, retain every pupil's assigned course. To keep it stable at each grade conditional on its older-grade assignments, rule out a proposed course if any contract it introduces would block: the applicant prefers it to their detailed assignment, and that seat type has a vacancy or would replace its lowest-priority holder. Apply both tests in every design draw. Courses the policies share, including continuity, stay fixed. These conditions are necessary and sufficient for the old detailed allocations to remain stagewise stable. For a target absent from the old menu, one added hit is possible precisely if some target-bearing proposed course passes and some old-only course can be removed. Maximizing each row separately and summing gives the exact across-draw hindsight ceiling, counting an April school once even if several courses have that school.

Does the mechanism select every old detailed allocation when run again? The stagewise ceiling is an upper bound for this stronger objective. Any exact replication of the old outcomes must make them stagewise stable under the same older-grade assignments. The reverse implication need not hold: DA in a higher grade can select a different stable allocation, changing which younger siblings receive priority. For a sufficient lower bound, keep every old-only course that occurs at or before the pupil's old partner in the operational order in any draw, and add a proposed-only course only if it follows the old partner in every draw. Treat being unplaced as last. This keeps each pupil's entire proposal sequence up through its last proposal intact, including all seat-type attempts. Induction on DA proposals within a grade and then on descending grades preserves the full detailed outcome. The same row calculation optimizes the label under this stronger sufficient condition.

*Why a conditional witness does not guarantee that every junior keeps a seat.* There are two grades, with one seat at each relevant program. In the higher grade $H$ ranks $x \succ u \succ y$ and changes the displayed pair from $(x, u)$ to $(u, y)$. $K$ lists $(x, y)$ and $G$ lists $(y, x)$. At $x$ the order is $G \succ H \succ K$, and at $y$ it is $K \succ G \succ H$. Initially DA gives $(H, K, G) = (u, y, x)$. That assignment remains stable on the new display: the only newly added course is below $H$'s assigned $u$. DA on the new lists nevertheless gives $(u, x, y)$, which $K$ and $G$ strictly prefer. In the junior grade, $B$, a sibling of $K$, lists $(x, y)$; $L$ lists only $x$. At $x$ the ordinary order puts $L$ ahead of $B$. With $K$ at $y$, $L$ gets $x$ and $B$ gets $y$. With $K$ at $x$, $B$ has sibling priority there, gets $x$, and leaves $L$ unplaced. Every old contract can survive in a stable allocation conditional on the original higher-grade assignments. Running applicant-proposing DA grade by grade changes those assignments and displaces $L$. The example uses only fixed reports, individual seats and the documented higher-grade sibling priority.

*Checking the menu the oracle selected.* Keeping every proposal prefix permits 152 additional hits (41, 50 and 61). To test the upper bound directly we constructed one common menu profile per cohort. For each additional pupil it adds the earliest eligible course at the April school and drops

the last removable old-only course. Every other pupil keeps the blind menu. Full re-clearing on each of the 1,024 matched design draws for each cohort reproduced every baseline course and seat type, including unplaced applicants: 1,580,726 × 1,024 = 1,618,663,424 individual comparisons and zero differences. Thus 314 attains the necessary upper bound even when the platform must reproduce the entire chosen assignment on these draws. We also obtained identical assignments at the three recorded orders and in 192 fresh matched-course draws. On 48 fresh draws for each alternative family model, the corresponding optimum menus changed no pupil's course under course-specific tickets; school-wide family tickets produced one exit in one draw. A finite panel cannot establish protection for every future order or a bound for a policy protecting only whether incumbents remain placed.

## G Testing responses to a five-course menu

*What the field study identifies.* Arteaga et al. [2022b] study a warning shown when a family tried to submit a list with high predicted nonplacement risk in 2018–20. The message encouraged adding schools without naming one. Tables II and III report the discontinuity at 30% risk. The probability of adding any school just below the threshold was .199; crossing it raised that probability by .216 (SE .010). The rise in appending was .205 (.009). The estimates refer to overlapping actions because a family could add multiple schools. The probability of placement at a listed school rose by .038 (.009). In their sample, the treatment induced 1.573 extra schools per induced adder (.090). We verified the coefficients against the archived tables [Arteaga et al., 2022a]. The assignment outcome for a warned family and the national placement difference between two recommenders are different estimands. In a separate Chilean experiment, access to an individualized report with suggested schools also changed list revision behavior [Agte et al., 2024]. Neither study observes our two menus.

*National simulations.* We use the two frozen menus, the original quota, the same four complete national cohorts, and the reconstructed allocation from Section 3. An addition can come from a course the family ranked on its original eligible report. In the school-level tests, its establishment must be absent from all courses on the displayed menu. The first candidate is the highest-ranked qualifying course. The alternative is the one with the lowest ratio of eligible reports plus one to published vacancies plus one. Submitted rank breaks ties. This proxy uses no enrollment or priority draw. It does not guarantee a seat. A course selected for a pupil is identical across the two ways of inserting it into that pupil's menu. We either insert it in its original rank or append it after the shown courses. Family promotions, sibling priorities, separate seat types and guaranteed returns are then processed afresh. A protected return can make a tail option unreachable; it still counts as a pursued school in the coverage calculation. A family with no qualifying option cannot add one in this test.

A common uniform draw for each pupil determines participation under both policies and across all rates. Intermediate rates use eight independently seeded repetitions in each cohort. Tables report the mean of the four-year sums. The variation across seeds measures uncertainty from this simulation design. At a response rate of .415, appending the first school gives 1,656 fewer linked placements on average (Monte Carlo SE 1.56); inserting it in submitted order gives 1,340.75 fewer (SE 4.01). To vary the number of additions, 57.3% of the respondents in the .415 scenario try to add a second distinct omitted school. This is a modeling choice motivated by the field study's estimate of 1.573 schools per induced adder. The actual average is 1.289 under either menu because many reports contain only one recoverable school. Tail insertion gives 2,817.6 more April hits and 1,645 fewer linked placements. Original-order insertion gives the same hits and 1,279 fewer linked placements. We also tested probabilities .10, .216 and .50. Under the append rule we crossed all

seven tested response rates (including zero and one) for the two menus. Linked placement fell in every draw at all 49 rate pairs, with differences from −1,694 to −1,608. Coverage can change sign when uptake differs by policy. If everyone adds the first available school under the blind menu and no one does under the informed menu, the informed rule has 1,386 fewer hits and 1,694 fewer linked placements.

For a targeted check, we estimated nonplacement risk on the two existing displays with 64 simulated course-specific family lotteries per cohort, before using the recorded order for the final allocations. The stochastic check uses 2022–24. In those years 50,571 applicants had estimated risk between 20% and 40% on both displays. Only 5,907 could add a previously unshown school on their blind menu and 5,917 on their informed menu. With .415 participation restricted to that group, the mean linked placement difference was −1,359.6 if they appended it, compared with −1,360 under binding menus. In a separate check, giving the observed same-grade returners the least-congested candidate to append gives 4,671 more April hits and 1,625 fewer linked placements. The simulated threshold group is selected by a model of the frozen displays; the threshold in the field study used a different forecast.

*Which families respond?* Let $G_A$ and $G_B$ be the linked pupils whose blind or informed menu, respectively, misses their historical April school and would gain that school by adding its chosen candidate. Let $S$ be any common set of pupils willing to expand. Write $\Delta_0$ for informed minus blind coverage before anyone expands. Then

$$\Delta(S) = \Delta_0 + |S \cap (G_B \setminus G_A)| - |S \cap (G_A \setminus G_B)|.$$

This follows by summing the change in each pupil's hit indicator. Membership in both gain sets cancels. The minimum is $\Delta_0 - |G_A \setminus G_B|$ and the maximum is $\Delta_0 + |G_B \setminus G_A|$. Either endpoint is achieved by taking the corresponding exclusive set. More generally, among $U$ eligible pupils let $b$ and $r$ be the blind-only and informed-only gain counts. For exactly $m$ willing pupils the sharp lower endpoint is $\Delta_0 - \min(m, b) + \max(0, m - U + r)$. The upper endpoint is $\Delta_0 + \min(m, r) - \max(0, m - U + b)$. Sorting the individual contributions $-1, 0, 1$ and taking the first or last $m$ proves both claims. For the first omitted school the two exclusive counts are 4,817 blind and 774 informed, with 1,207 shared. For the least-congested option they are 3,718 blind and 1,256 informed, with 995 shared. The endpoints again hold when willingness is fixed at 19.9% of applicants able to add under either menu, rounded down in each cohort. We supplement each exclusive group with pupils who are in neither exclusive group. This gives 34,742 willing applicant-years in each construction, out of 174,594 eligible. Enrollment labels enter these deliberate worst- and best-case constructions. They never enter the predictors or the ordinary response simulations.

With policy-specific marginal response probabilities $q_A, q_B$ common among each policy's potential gainers, expected coverage for the first-school rule is $4{,}638 + 1{,}981q_B - 6{,}024q_A$. No independence across applicants is needed for this expectation. Under equal probabilities the expected gap is positive even at full response. For placements we clear the entire market for each concrete responder set. These sharp coverage bounds do not bound the simulated placement gap.

*Complementary-round choices and seats.* The repeat-course comparison links each same-grade B2 applicant to their B1 record, excludes automatic continuity entries, and matches every explicitly ranked C2 school/course to the eligible original report. It adds every match that the pupil's policy omitted. The original-rank version yields 4,733 additional hits and 1,651 fewer linked placements. Appending in the complementary order yields the same hits and 1,653 fewer placements. Restricting the additions to the group with no historical regular offer through the wait list leaves the original linked placement gap of 1,670 under either insertion order. Among the 2,635 pupils who lose a binding-menu place in the full cohort, 408 subsequently reapplied at the same grade in the historical

data, 390 explicitly named a new school and 122 subsequently enrolled at a newly listed school that had offered them an explicitly requested course. These counts describe their observed historical paths.

For the separate second-stage calculation let $U_A, U_B$ be the pupils unplaced under blind and informed first-stage menus. For a set $U$ define $r(U)$ as the maximum number that can be matched to the recorded second-stage schools using common capacities. Construct a unit-capacity edge from a source to each eligible pupil and from that pupil to each explicitly applied-to course; each course has an edge to the sink of capacity equal to its vacancies. Integral maximum flow gives the count. A minimum cut certifies each value. No first-stage recipient moves. For the linked estimates the graph uses only linked pupils and therefore gives them first claim on second-stage seats. Both policies use the same priority-free objective and complementary inventory. Later-only applicants and the effects of earlier hypothetical placements on released vacancies are omitted.

Even with other competitors in common, a useful bound holds for maximizing headcount. Restrict any matching on $U_B$ to $U_A \cap U_B$. It leaves a feasible matching on $U_A$, and the removed edges can cover at most $r(U_B \setminus U_A)$ pupils. Hence $r(U_B) - r(U_A) \leq r(U_B \setminus U_A)$. The same proof allows any set of extra common applicants with fixed options. On the new-school graph with published capacities this upper bound is 388 for all applicants and 358 for the linked-only graph. Closing the original deficits of 1,834 or 1,670 therefore requires some additional change to the graph, supply or objective. This inequality compares maxima, conditional on who can pursue which seats; two arbitrary priority-based assignments need not satisfy it.

Table 15. Rank-informed minus rank-blind after a hypothetical second chance. Each arm keeps its initial recipients and maximizes placements among its unplaced pupils on observed second-round choices. Linked placement gives linked pupils first claim. A pursued hit includes a second-round choice only for a pupil unplaced by that policy and only at a course with positive common capacity.

| Courses allowed | Common supply | All placed | Linked placed | Pursued hits |
|---|---|---|---|---|
| New school | Published second round | −1,558 | −1,419 | +4,752 |
| All explicitly listed | Published second round | −1,550 | −1,409 | +4,763 |
| New school | Also cap at both residuals | −1,580 | −1,439 | +4,743 |
| All explicitly listed | Also cap at both residuals | −1,574 | −1,433 | +4,748 |

The residual supply check caps each course by its available regular-round seats after each of the two initial allocations as well as its published complementary vacancies. A course absent from the regular inventory receives zero in this check. All four rows relax category restrictions and competition from other second-round applicants. The relation to an actual two-round allocation depends on which families would reapply, list new schools, release old places and accept a subsequent offer.

*Why adding a backup can leave fewer pupils placed.* With fixed priorities, appending options can only increase a program's DA occupancy. Resume the old run with the extra proposals. A program either fills a vacancy or replaces its current lowest-priority holder. Chile's higher-grade sibling priority changes this argument. For any $k \geq 2$, appending one course for one applicant can reduce total placement by $k - 1$.

To see this, suppose $H$ and $S$ each list only $u$ in the senior grade. There is one seat at $u$, which admits $H$, and a vacant senior seat at $x$. In the junior grade $k$ pupils list only $x$, and each of $S$'s $k$ younger siblings ranks $x$ above $y$. Both junior courses have $k$ seats. In the ordinary priority order at $x$, the pupils listing only $x$ outrank every sibling. They get $x$ and the siblings get $y$. Now append senior $x$ to $S$'s list. $S$ obtains that previously empty seat, giving all the younger siblings priority at

junior $x$. They take its $k$ seats. The pupils listing only $x$ become unplaced and nobody takes junior $y$. The senior grade gains one recipient and the junior grade loses $k$. This example uses only regular seats, with no family preference promotion. The institutional solver reproduces it already at $k = 2$, with six applicants and a fall from five placed to four.

Table 16. How the assumed response changes the comparison between recommenders. Entries are rank-informed minus rank-blind in Chile, and Haiku minus uniform in New York. Chilean totals use the fixed-priority pooled model over four application years. NYC uses all renters with the original geographic weights. In the last row of each panel both policies list the same eligible options.

| | Placements | First choices | Past offers on list |
|---|---|---|---|
| *Chile, changes in pupils* | | | |
| Pursue the five shown | −1, 787 | +2, 218 | +5, 552 |
| Restore one at its original rank | −940 | −846 | +676 |
| Append that same option last | −1, 761 | +2, 163 | +676 |
| Append every omitted option | −1, 331 | +2, 005 | 0 |
| | Ordinary placed | Enriched placed | Enriched first choices |
| *NYC, changes in percentage points* | | | |
| Pursue the five shown | −2.71 | −5.40 | +2.87 |
| Restore one at its original rank | −1.98 | −4.28 | +0.18 |
| Append that same option last | −2.17 | −4.41 | +2.97 |
| Append every omitted option | −0.38 | +0.28 | +1.84 |

*New York and the pooled Chilean benchmark.* In this earlier pooled comparison, to restore one option, take each applicant's highest originally ranked eligible option missing from that policy's menu, if one exists. For a given policy the recovered option is the same in both insertion scenarios. This change alone flips the first-choice comparison in *each* Chilean cohort. Across the four, adding the course in its reported position gives 846 fewer first choices, while putting it last gives 2,163 more. The option sets and the +676 historical-offer coverage difference are identical across these two scenarios. In the in-order scenario the blind rule gives both more placements and more first choices. It therefore wins for every nonnegative first-choice bonus in the index from Section B. A returning pupil cannot reach an option below their protected course in the replay. Inserting the extra option immediately before that entitlement instead of after it gives −1, 764 placements and +2, 161 first choices across the cohorts.

The ordinary NYC first-choice differences for in-order restoration and last-place insertion are −0.13 and +2.17 points. The former interval spans zero ($[-0.47, 0.22]$), as does the corresponding enriched interval ($[-0.16, 0.54]$). If everyone tries all omitted options after the shortlist, the enriched placement difference estimates +0.28 points with a paired simulation-block interval of $[-0.07, 0.65]$. The ordinary estimate is −0.38 with interval $[-0.66, -0.11]$. Access to the full set therefore substantially changes the housing result. If all applicants know their options and can rank them freely, truthful reporting is weakly dominant in the applicant-proposing DA model [Dubins and Freedman, 1981]. Restoring the full original order then makes the two rules identical. Trying the recommendations first is a behavioral assumption about how families use advice.

We also allowed each applicant, independently, to pursue their entire original list with probability one-half and otherwise use the five shown. The same draw applies to the person under both policies. The Chilean difference averaged 887 fewer placements (Monte Carlo SE 1.69) and 1,057 more first choices (SE 2.46). The NYC placement differences were −1.03 and −1.63 points for ordinary and

enriched inventory. These probabilities describe a sensitivity analysis. Actual decisions to keep looking could depend on an applicant's preferences, information and expected chances.

## H Choosing five courses with complementary admission chances

Ali and Shorrer [2025] study how an applicant can hedge when admission decisions are correlated. Arteaga et al. [2022b] offered Chilean families advice about their risk of going unplaced. Our test fixes the five discretionary slots, uses only the submitted courses and re-clears everyone together.

Use 64 draws of the course priorities within their public tiers. Clear the full reports once per draw. A course is attainable at that draw's cutoff if it has a free seat or would retain the pupil at the priority of the lowest-ranked holder. For each pupil choose the menu covering the most distinct draws through at least one attainable course, including the protected option when present. Among ties maximize the number of simulated full-list partners shown, then the total number of attainable pairs summed over draws, then prefer earlier submitted ranks. We solve the resulting five-choice problem exactly by enumerating subsets when the simpler tie-breaking rule falls short of the full union. A separate oracle checked 2,048 small instances. The full-list cutoffs stay fixed only while choosing menus. To evaluate, run DA again on the selected courses under the recorded priorities.

Table 17. Joint cutoff coverage versus selecting the five most frequent simulated partners. Each entry is a separate median of paired counts over eight batches, each using the same recorded priorities for evaluation. April agreement counts replay placements at the school where the pupil historically enrolled.

| Year | Places | Places vs. tie change alone | First choices | April schools shown | April agreement |
|---|---|---|---|---|---|
| 2022 | +42.5 | +24.5 | −55 | +151 | −114.5 |
| 2023 | +39.5 | +21.5 | −46.5 | +137 | −83 |
| 2024 | +40.5 | +22 | −40.5 | +254 | −61 |

All 24 independent-course batches gained between 27 and 57 placements over partner frequency. Merely resolving frequency ties in favor of courses attainable in many draws accounts for part of this gain. The further gain from jointly covering draws ranged from 9 to 30. The original rank-blind predictor placed a median 387.5, 435.5 and 426 more pupils than the joint rule in the respective years. Gross gains and losses can be much larger than the net. In 2022 a median 175 pupils gained seats and 133.5 lost seats relative to partner frequency. Among pupils without an initial offer the net ranged from −2 to +41, +1 to +23, and −13 to +9 across years.

For a sensitivity check we instead drew one common applicant order per simulation and used it at every course within tiers. We trained all comparison policies on the same draws and evaluated them at the same recorded priorities as before. Joint coverage gained a median 94, 125 and 118.5 placements over frequency, respectively, yet its extra gain over changing only the tie-break ranged from −4 to +2 in all 24 batches. Among pupils without an initial offer the median placement changes against frequency were −92, −88 and −74. These repetitions measure algorithm sensitivity in the same three cohorts.

*Why the cutoff problem can be hard.* Let $S_j$ be an arbitrary subset of a finite list of scenarios. Give each course $j$ one seat and one applicant $a_j$ who lists only that course. One other pupil $f$ lists every course. In scenario $d$, use a common priority order that puts $a_j$ above $f$ if $d \notin S_j$ and below $f$ otherwise. In full-list DA, $f$ gets the first course where they outrank its applicant, if any. Every other course takes its own applicant. Thus $f$ meets the cutoff at $j$ exactly on $S_j$. If $f$ keeps only a menu $M$, they actually receive a seat on precisely $\bigcup_{j \in M} S_j$, yet the number of occupied seats is always the number of courses. Choosing $k$ options therefore realizes the classical maximum coverage

problem [Feige, 1998]. Asking if all scenarios can be covered is Set Cover and is NP-complete when $k$ and the explicitly listed scenario count vary. A candidate menu is checked by replaying DA. This construction permits arbitrary scenario support. Our empirical budget of five admits exact enumeration.

## I What changes during a partial rollout?

A service might try the new rule on a fraction of applicants and use the result to judge a full switch. We simulated that decision on the original 480 rental markets and two school cohorts. Independently for each applicant, we use policy $B$'s full menu with probability $p$ and otherwise keep $A$'s. We clear the entire market at each $p \in \{0, 1/4, 1/2, 3/4, 1\}$. We use 64 independent pairs of random assignments, with complementary uniforms within a pair and shared draws across uptake levels. This design gives a Monte Carlo standard error conditional on the fixed cohorts, preferences and candidate menus. For NYC we also resample whole simulation seed blocks. These experiments were specified after the original full-switch comparisons.

For any fixed reference offer, the expected number of people shown that offer is exactly the interpolation between the two endpoint counts: menu membership depends only on that person's switch. At half adoption the complementary assignments give that midpoint exactly in each pair. Placement need not follow the same line. For example, if two one-slot menus both show $x$ before the switch and both show $y$ afterward, either endpoint places one applicant. A mixed market with one applicant at each site places two. Reversing two initially distinct menus can bend the placement curve in the opposite direction.

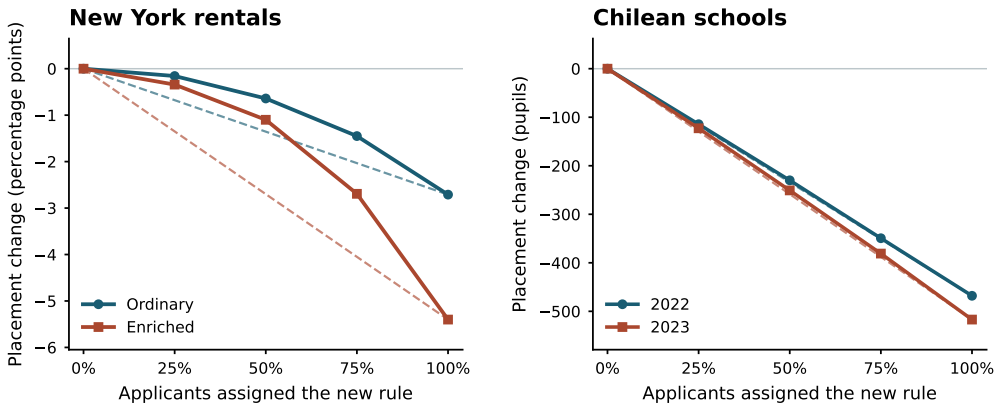



Fig. 2. Placement change from the original policy when applicants independently switch to the proposed menu. Points give the simulated mean at each uptake. Dashed lines interpolate between full adoption and the original policy. The Chile panel reports pupils within each national cohort; the NYC panel uses geographically standardized percentage points.

Placement at half adoption exceeded the interpolation by 0.71 and 1.60 percentage points in New York. The corresponding 95% simulation-block intervals were $[0.59, 0.84]$ and $[1.40, 1.82]$, with Monte Carlo standard errors of 0.009 and 0.010 points. In Chile the midpoint deviation was about four pupils in 2022 and eight in 2023 (Monte Carlo standard errors 0.45 and 0.44 pupils, respectively). The school curves are close to linear in aggregate, yet applicants still change places. At half adoption an average of 272 pupils in 2022 and 271 in 2023 lost a place on a menu identical

under both policies. The corresponding gains were 145 and 107. A nearly linear aggregate curve offers little protection to an individual family.

## J How many vacancies could one referral fill?

With tentative offers in hand, could the platform fill a vacancy by referring one unmatched applicant to it? We consider only affordable, feasible apartments or eligible courses already on the original report. Each referral must preserve the existing contracts together in a stable allocation.

### J.1 One coordinated referral after offers

Suppose the platform now observes an assignment $M_0$ stable on the initial lists. Join each unmatched applicant to each mutually acceptable program with a current vacancy. These edges are hidden: if one were displayed it would block $M_0$. Give each program its remaining capacity and let $r$ be the size of a maximum matching in this graph. With lists of $\min(k, d_i)$ programs, where $k \geq 1$ and $d_i$ is feasible degree, a hidden option can always replace a displayed one. If an entitlement is protected outside the quota, the substitution is made in a discretionary slot. In both replays, every selected applicant had a full five-slot discretionary list.

Theorem J.1. *Among extensions that keep every initial contract in a single stable assignment, the maximum number of additional recipients is $r$. One shortlist substitution per selected applicant attains it. Every stable outcome on the revised display gives each consenting applicant their named referral, even if others decline. Under applicant-proposing DA, every original recipient also receives a weakly preferred assignment.*

Proof. The new contracts in any extension must form a residual matching. Conversely, choose any such matching $A$. For each consenting participant substitute their assigned program for their last discretionary displayed option. Write $W$ for the original assignment together with the accepted referrals. All new pairs in the display belong to $W$. On old edges the applicant weakly improved. An originally full program has the same holders. At an originally slack one, every applicant with an old edge weakly preferred their original assignment, by stability. Hence $W$ is stable on this display. DA weakly dominates $W$ for applicants. All stable selections have the same matched people [Roth, 1986]. If some stable selection omitted an accepted referral $e$, it would keep its recipient matched even after deleting that unused edge. On that same graph $W \setminus \{e\}$ is stable by the preceding argument, since this applicant now has a subset of their original list. That would leave the person unmatched in one stable outcome only, a contradiction. □

The maximum fixes the original contracts in a stable witness. Opening new options to incumbents would be a different intervention. The empirical calculation also requires deciding which hidden options a person would actually accept. We report both a top-five rank limit and all mutually acceptable vacancies. Work on facilitating matches, adding school seats and guaranteed school referrals provides related interventions [Gajulapalli et al., 2020, Lepe, 2025, Trabelsi et al., 2025].

In ordinary NYC markets only 37 unplaced lower-income renter instances could use any vacancy. Of the maximum 683 additional placements, at least 22 and at most 25 could go to them. In enriched markets 1,601 unplaced lower-income renters could use a vacancy. Maximizing total referrals first, then minimizing or maximizing referrals to this group, gave 1,002–1,085 lower-income recipients among 1,926 placements. Both supply and the households sampled change across these designs.

Many residual school seats were in courses that no unmatched pupil had listed. With full take-up, at most 224 placements could be added across the two years. Just one was possible within the family's top five. Neither year had an eligible referral after the rank-blind allocation, so even after

Table 18. Largest number of named referrals compatible with retaining all existing contracts in one stable allocation. NYC uses simulated Haiku offers. Chile uses rank-informed pooled DA offers. All counts use instances in their own row. A vacancy must be at an affordable, feasible listing or an eligible course already on the pupil's reported list.

| | Vacant seats | Unplaced with an option | Any rank | Top five |
|---|---|---|---|---|
| NYC ordinary | 7,972 | 757 | 683 | 204 |
| NYC enriched | 5,990 | 2,527 | 1,926 | 370 |
| Chile 2022 | 601,565 | 128 | 127 | 0 |
| Chile 2023 | 640,751 | 97 | 97 | 1 |

each policy's largest extension the rank-informed policy placed 761 fewer pupils. After uniform rental menus the corresponding maxima were 215 ordinary and 518 enriched placements.

We verified the school maxima with a second national flow implementation. We replaced each selected applicant's last discretionary option with their referral, reran DA, and checked each named assignment, the weak improvement for earlier recipients, the quota and continuity rights. A service would need to observe current offers, elicit acceptable alternatives and give each family a chance to decline. Table 18 reports the opportunities available under those conditions.

### J.2 What if a current recipient would like to move?

A vacancy can help someone who cannot use it directly. Suppose that Ana and Ben both see only course $x$, which gives Ana priority. Ben goes unplaced. If Ana ranks the vacant course $y$ above $x$, showing her $y$ lets her move there and Ben take $x$. Ben needs no change to his own shortlist. More generally, can we find such trades without making another applicant worse off?

Fix a stable assignment $M$ on the old display $D$, strict applicant preferences and strict, responsive program priorities. For program $j$, its *claimants* are the applicants with $j \in D_i$ who strictly prefer it to $M(i)$. An unmatched applicant prefers every acceptable option to being unmatched. A program with any claimants must already be full and give every incumbent higher priority than every claimant. Write $H_i$ for the hidden options permitted by the intervention. We use either all hidden acceptable options or only those in the applicant's original top five. Keep each baseline contract. Among $D_i \cup H_i$, admit a possible move $(i, j)$ if $i$ strictly prefers $j$ to $M(i)$ and $j$ ranks $i$ at least as high as its highest-priority claimant. A program without claimants imposes no priority restriction. In the quota model, a hidden option may enter if the applicant has a replaceable discretionary slot. Protected options always remain visible.

Theorem J.2. *Let $L$ be the largest number of additional placements in the resulting graph subject to keeping every incumbent matched and each program at least as full as before. A single maximum flow finds $L$ and a collection of vacancy chains, each adding one person. For any subcollection of whole chains, at most one substitution per participating applicant implements a stable witness on the new display. Applicant-proposing DA places exactly the witness's recipients and gives each a weakly preferred assignment. A second maximum flow on the weak improvements in $D_i \cup H_i$, dropping the priority and slot restrictions, gives an upper bound $U$ on additional placements in any assignment weakly improving every applicant on these pairs at the same capacities.*

Proof. Start with the flow representing $M$. In its residual network, connect the source to each unmatched applicant, direct available contracts from applicants to programs, reverse each held contract, and connect each program to the sink with its residual capacity. Integral augmentations preserve all current recipients and never reduce program occupancy. The omitted arcs into the

source and out of the sink cannot occur on a simple augmenting path. Thus max-flow/min-cut gives the optimum. Discard flow cycles and decompose the remainder into paths. Each applicant lies on at most one path: an unmatched applicant has one unit of inflow from the source, and a matched applicant has one unit from the reverse of their old contract. A program may occur on several paths. Any subset of paths preserves the same inequalities.

Call the resulting assignment $W$. For each hidden witness contract, substitute its program for a displayed discretionary option. An old edge preferred to $W(i)$ was already preferred to $M(i)$, so its applicant was a claimant. That program remains full. Every new holder ranks at least as high as the best old claimant. Equality can occur only if the holder is that claimant, who cannot then block their own contract. Every added edge belongs to $W$, so $W$ is stable. Applicant optimality and invariance of the matched set yield the DA conclusion [Gale and Shapley, 1962, Roth, 1986]. Finally, every allocation weakly improving on $M$ uses only applicant-approved edges of the second flow graph, or baseline contracts. A maximum ordinary flow there bounds its size; augmenting from $M$ attains that graph's maximum without reducing occupancy or unmatching incumbents. □

Table 19. Additional placements after tentative offers. Direct referrals fix all old contracts. Chains may move incumbents to options they prefer. The ceiling drops program priorities. The right three columns restrict newly displayed options to each person's original top five.

| | All acceptable ranks | | | Top five only | | |
|---|---|---|---|---|---|---|
| Starting policy | Direct | Chains | Ceiling | Direct | Chains | Ceiling |
| NYC ordinary, uniform | 215 | 641 | 1,006 | 35 | 411 | 629 |
| NYC ordinary, Haiku | 683 | 983 | 1,353 | 204 | 610 | 843 |
| NYC enriched, uniform | 518 | 1,646 | 2,102 | 49 | 499 | 577 |
| NYC enriched, Haiku | 1,926 | 2,493 | 2,844 | 370 | 804 | 901 |
| Chile, rank-blind | 0 | 53 | 133 | 0 | 50 | 127 |
| Chile, rank-informed | 409 | 475 | 649 | 1 | 2 | 4 |

In Chile, the rank-informed menus began 1,787 placements behind the rank-blind menus across four cohorts. Even ignoring priorities, giving every incumbent a course at least as preferred as their original assignment leaves a shortfall of at least $1,787 - 649 = 1,138$ placements relative to the unchanged rank-blind policy. Requiring newly shown options to be in a pupil's original top five lowers the ceiling from 649 to four. Chile's applicant-proposing replay uses reported courses, pooled capacities, fixed priorities and protected continuity options. The bound concerns this reported option set.

In the rental simulations the corresponding answer changes with the options people accept. With all modeled acceptable pairs available, chains after Haiku reach or exceed unchanged uniform headcount in 226 of 240 ordinary markets and all 240 enriched markets. In the enriched design, requiring newly shown units to be in the renter's top five gives 804 additional placements against an original deficit of 898. Even relaxing the priority constraint yields only 901. When all acceptable options may enter, applying chains to both starting policies leaves Haiku 128 placements behind uniform in ordinary markets and 51 behind in enriched markets. Each design pools 64 simulated renters in each of 240 markets. These are offer counts under modeled rental preferences; neither the identity of the original uniform recipients nor a signed lease follows from recovering the headcount.

Existing mechanisms reassign seats after cancellations [Feigenbaum et al., 2020], study entry in a second round [Gajulapalli et al., 2020], or place additional capacity where it improves assignments [Bobbio et al., 2026]. Guaranteed referrals to new schools have been studied in Chile [Lepe, 2025],

and allocation facilitators can optimize changes to a compatibility graph [Trabelsi et al., 2025]. Here the seats, acceptable options and realized priorities stay fixed. The calculation asks how many already acceptable contracts a list of unchanged size can recover under DA, with an explicit bound from applicants' willingness to move. Using a chain requires coordinated take-up by its movers. Selective display can remove an unchosen course on which someone previously had a priority claim. Stability here is assessed on the final displayed lists. The welfare guarantee uses the original ranking of the assigned courses. The theorem gives no incentive guarantee for preference reports.

## K School-policy benchmarks and institutional sensitivity

*A predictor that simulates the allocation.* For 2022 and 2023 we used completed reports and priority tiers while withholding the within-tier order. Simulating DA to predict admission risk has precedent in Arteaga et al. [2022b, online appendix, Sec. F.1]. Here we hold all submitted lists fixed, draw an independent uniform order within each course and tier, and run DA on the full lists. Repeat 64 times. For each pupil keep the five discretionary programs they received most often, breaking frequency ties by submitted order and retaining the entitlement. We evaluate the resulting menus at the published order. This benchmark requires completed reports and uses no historical offers in selection.

Across eight independent 64-draw batches per year, this rule beat the rank-informed predictor on both recorded-offer coverage and placement in every comparison. It showed 494–541 more historical offers and placed 25–56 more pupils, but gave 106–163 fewer pupils their first choice. Compared instead with the rank-blind predictor, it still covered more historical offers and placed fewer people in every batch. Simulation precision matters: at 16 draws, only five of the 16 comparisons improved on rank-informed on both outcomes. In the first 64-draw batch there was still a zero-frequency tie at the fifth-slot boundary for 32,982 of 42,123 longer lists in 2022 and 35,800 of 43,409 in 2023. The fallback to submitted rank is part of the rule. The eight batches vary the estimated menus within each fixed cohort.

We then fixed the comparison, including all three simulation lengths and eight batches, before generating the new rule's 2024 outcomes. The cohort and the original predictors' outcomes were already known. At 64 draws the historical coverage advantage persisted in all eight batches (+447 to +483 offers), as did a much smaller placement advantage (+2 to +24 pupils). First choices fell by 105–123. At 16 draws *none* of the eight batches placed more pupils; at 32 only three did. We did not extend the simulated-lottery test to 2025, the year the ministry replaced random tie-breaking.

Pooling all seats simplifies the ministry's allocation. A sensitivity that first honors continuity and gives designated pupils priority for earmarked vacancies changes the year-specific net placement losses to 469 and 516. Neither version models all reserves or dynamic family priorities; overlapping reserve categories in particular can change who is admitted [Doğan et al., 2026]. We also tested a deliberately generous seat count, $v_j + h_j$, to allow for the possibility that vacancies exclude continuing pupils. This adds 174,270–225,476 seats per year and reduces full-list agreement with historical outcomes to 82.01–85.68%. With the original menus frozen the placement differences remain negative in all years. Recomputing demand-to-capacity features and choosing new menus under the same 2019 coefficients yields 337–498 fewer placements and 839–1,593 more historical offers shown per year. The two seat conventions are scenarios; neither recovers the ministry's full mechanism. Appendix F compares both predictors under the reconstructed rules. Only the two frozen predictors' historical-offer and placement directions were specified before acquiring the validation cohorts.

*List length.* Keeping both fitted scores fixed, we repeated the comparison at one, three, five, eight and ten discretionary slots in 2022 and 2023. In all ten year–quota combinations rank-informed

covered more historical offers and placed fewer pupils in replay. The size varied sharply. At one slot, historical coverage rose by 37,861 and 42,580, while placement fell by 18,907 and 17,922. At ten slots, coverage rose by only 34 and 70 and placement fell by 48 and 51. Once all reported options fit, the policies give the same allocation by construction.

At the five-slot quota we also varied how ties are resolved within the original priority tiers in the 2022–23 replays. We drew new independent course-specific lotteries and, separately, a common applicant lottery across courses. We used 32 paired draws per ordering scheme and year, recalculating the full-list reference for each draw. The rank-informed policy raised full-partner coverage and lowered placement in every one of the 128 paired comparisons. Net placement differences ranged from −557 to −393 pupils. Historical-offer visibility stays fixed when we change only the allocation order, so it supplies no additional test in this exercise.

## L AI-assisted research and writing

The authors proposed the intermediation question, built the original rental pipeline and identified the shortage of affordable listings in the pilot. They used OpenAI Codex (GPT-6 family) extensively to develop and challenge proofs, write and check the national replays, analyze administrative records, and draft and revise the text. The record identifies the model family but does not preserve a checkpoint for every earlier interaction. One request was "Can we test whether the prediction-versus-placement tradeoff holds on a real matching dataset with observed preferences, then fold that evidence into the paper?" A later request asked for a reviewer audit of the theory, experiments and connection between the two settings. The resulting revisions distinguished historical enrollment from simulated placements, tested different comparators and application orders, and checked numerical claims using separate implementations. The rental experiment used the saved selections of the separate model identified as `claude-haiku-5-5`. Its inputs and cache limitations appear in Sections 8 and 10.

The subsequent design work included AI-assisted construction of the finite-lottery reduction, vacancy-chain algorithm, cutoff comparison and their counterexamples. The published code records the simulated priority inputs, independent small-instance checks, control policies and aggregate outputs.

The institutional revision used AI assistance to reconstruct seat categories and family rules, compare predictions against the initial-offer file, and rerun the menu optimization and assignment checks. The release includes the comparison code and aggregate results.

The behavioral extension used AI assistance to link complementary applications to the shortlist comparison, check published field estimates, construct response scenarios and sharp coverage bounds, and verify the national re-clears and second-stage capacity calculations.